



Review

Editor's Choice

Urban Heat Island Effect: Remote Sensing Monitoring and Assessment—Methods, Applications, and Future Directions

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Review

Urban Heat Island Effect: Remote Sensing Monitoring and Assessment—Methods, Applications, and Future Directions

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Abstract

This study systematically reviews the development and application of remote sensing technology in monitoring and evaluating urban heat island (UHI) effects. The urban heat island effect, characterized by significantly higher temperatures in urban areas compared to surrounding rural regions, has become a widespread environmental issue globally, with impacts spanning public health, energy consumption, ecosystems, and social equity. The paper first analyzes the formation mechanisms and impacts of urban heat islands, then traces the evolution of remote sensing technology from early traditional platforms such as Landsat and NOAA-AVHRR to modern next-generation systems, including the Sentinel series and ECOSTRESS, emphasizing improvements in spatial and temporal resolution and their application value. At the methodological level, the study systematically evaluates core algorithms for land surface temperature extraction and heat island intensity calculation, compares innovative developments in multi-source remote sensing data integration and fusion techniques, and establishes a framework for accuracy assessment and validation. Through analyzing the heat island differences between metropolitan areas and small-medium cities, the relationship between urban morphology and thermal environment, and regional specificity and global universal patterns, this study revealed that the proportion of impervious surfaces is the primary driving factor of heat island intensity while simultaneously finding that vegetation cover exhibits significant cooling effects under suitable conditions, with the intensity varying significantly depending on vegetation types, management levels, and climatic conditions. In terms of applications, the paper elaborates on the practical value of remote sensing technology in identifying thermally vulnerable areas, green space planning, urban material optimization, and decision support for UHI mitigation. Finally, in light of current technological limitations, the study anticipates the application prospects of artificial intelligence and emerging analytical methods, as well as trends in urban heat island monitoring against the backdrop of climate change. The research findings not only enrich the theoretical framework of urban climatology but also provide a scientific basis for urban planners, contributing to the development of more effective UHI mitigation strategies and enhanced urban climate resilience.



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Keywords: urban heat island effect; remote sensing monitoring; land surface temperature; heat island intensity; multi-source data fusion; thermally vulnerable areas; green space planning; urban morphology; climate change; artificial intelligence

1. Introduction

1.1. Urban Heat Island Concept, Formation Mechanisms, and Impacts

Urban heat island effects can be classified into four main types based on different measurement methods and spatial scales, each possessing unique physical characteristics and observation approaches. The surface urban heat island (SUHI) metric measures surface temperature through thermal infrared remote sensing technology, reflecting temperature differences between urban surface materials and surrounding environments. Canopy Layer Urban Heat Island (CUHI) is based on air temperature observations at building canopy height (typically 1.5–2 m above ground), representing direct thermal sensation in human activity areas; Boundary Layer Urban Heat Island (BUHI) is obtained through upper-air meteorological observations, reflecting the overall thermal impact of cities on the atmospheric boundary layer; Ground-based Urban Heat Island (GUHI) is measured through underground temperature sensors, characterizing changes in subsurface thermal environments due to urbanization.

This review specifically focuses on the surface urban heat island (SUHI) phenomenon, a choice based on three core rationales. First, SUHI has natural compatibility with satellite remote sensing technology, as thermal infrared sensors can directly acquire surface temperature data, providing a technical foundation for large-scale, continuous monitoring, while other types of heat islands often rely on sparsely distributed ground meteorological station networks, making spatially continuous observation difficult to achieve. Second, remote sensing technology enables multi-scale SUHI monitoring from global to urban block levels, providing complete spatial distribution information and overcoming the limitations of point-based observations. Finally, SUHI data can directly support decision-making in urban planning regarding land use optimization, green space layout, and building material selection, providing scientific evidence for the spatially precise implementation of heat island mitigation measures.

It should be noted that important differences exist between SUHI and CUHI. Surface temperatures are typically higher than air temperatures, and their spatiotemporal variation patterns are not entirely consistent. Wang et al. [1] utilized MODIS four-times-daily observation data to analyze canopy–surface heat island differences caused by air advection; Wang et al. [2] observed that night-time heat islands in Phoenix were not only larger in area (410 km² vs. 176 km² for daytime) but also showed maximum intensity increases of up to 5.35 °C, revealing significant diurnal variation characteristics of heat island effects. Despite these differences, SUHI serves as an important indicator of urban thermal environments and, through continuous observation via remote sensing technology, provides a unique and valuable perspective for understanding the impact of urbanization on thermal environments.

The urban heat island (UHI) effect refers to the phenomenon where urban areas exhibit significantly higher temperatures than surrounding rural areas; it is named after its temperature distribution pattern resembling a geographical “island” surrounded by water. This phenomenon has become a widespread environmental issue due to rapid global urbanization. Empirical studies show that urban temperatures typically exceed those of surrounding rural areas by 1–7 °C and, in extreme cases, by more than 9 °C [3]. This urban–rural temperature difference varies markedly across global regions and climate zones, ranging from moderate (0–5 °C) to significant (7–8 °C), and demonstrates a continuous growth trend, with global heat island intensity increasing by an average of 0.156 °C per decade [4].

The formation mechanisms of the urban heat island effect involve complex interactions among multiple physical processes. Urban building materials such as concrete and asphalt possess high heat capacity and thermal conductivity, absorbing and storing

large amounts of solar radiation and releasing heat slowly at night. The unique geometric structure of cities alters local airflow patterns and radiation balance, with the urban canyon effect between tall buildings significantly impeding heat dissipation. The expansion of urban impervious surfaces simultaneously reduces vegetation cover, decreasing regional evapotranspiration cooling capacity. Additionally, anthropogenic heat release from human activities (such as transportation, industrial activities, and building energy consumption) directly increases urban heat input. Urban air pollutants form a “greenhouse dome” that further enhances thermal radiation retention and reduces night-time longwave radiation dissipation. Comprehensive studies consistently demonstrate that the proportion of impervious surfaces is the primary driving factor of heat island intensity, explaining over 60% of land surface temperature variation in multiple studies. In contrast, vegetation exhibits varying degrees of cooling effects in most urban environments, with negative correlations with land surface temperature ranging from -0.14 to -0.66 [5]. This wide range of variation reflects the compound influence of vegetation coverage, vegetation types, climatic conditions, irrigation levels, and seasonal factors.

Urban heat island research carries both significant scientific meaning and social value. Scientifically, it represents a typical case for studying the impact of human activities on local climate, providing a key perspective for understanding human–urban–climate interactions. Socially, the heat island effect has extensive and profound impacts. For public health, the heat island effect significantly increases the risk of heat-related illnesses and mortality, with studies showing that mortality rates during heat waves can increase by 5–10% due to the heat island effect [6]. Regarding energy consumption, heat islands lead to surges in summer air conditioning use, increasing not only energy consumption but also forming a positive feedback loop through the emission of heat and greenhouse gases. At the ecosystem level, heat islands alter urban microclimatic conditions, affecting species composition and biodiversity. In terms of social equity, research has found that low-income communities often suffer more severe heat island effects, revealing environmental justice issues [7].

Against the dual background of intensifying climate change and continued global urbanization, systematically understanding and addressing the urban heat island effect has become particularly urgent. Moore et al. [8] noted that nearly half the world’s population now lives in urban settlements, with rapid urban growth often associated with environmental degradation and health risks, meaning that the scope of heat island effects and the affected population will continue to expand, posing greater challenges to urban residents’ health, environmental sustainability, and social equity.

1.2. Role and Advantages of Remote Sensing Technology in Heat Island Monitoring

1.2.1. Spatial Coverage Advantages and Observation Density

Remote sensing technology, with its unique spatial coverage advantages and multi-temporal observation capabilities, has become a core tool for urban heat island research. Unlike traditional meteorological station networks that can only provide discrete point data, remote sensing technology can present a complete spatial distribution picture of urban thermal environments. A single Landsat scene can cover an area of 185×180 km (approximately 33,300 square kilometers), containing over 37 million temperature observation points at 30-m pixels, while ground meteorological station networks in equivalent areas are often extremely sparse. Sub-Saharan Africa averages only one meteorological station per 1000 square kilometers, and the number of reporting weather stations in South America has dramatically declined from 4267 to 390 (a reduction of 91%). Particularly in developing regions, the number of real-time reporting stations is very limited, and traditional meteorological stations face numerous maintenance and operational difficulties, including a lack of technical personnel and high annual operating costs of USD 200–500 per

station [9]. This high-density observation advantage enables spatial pattern analysis of heat islands and precise identification of hotspots. Meanwhile, the regular revisit mechanism of satellite platforms provides a reliable data foundation for tracking temporal dynamics and analyzing long-term evolution trends of heat island phenomena. The Landsat series, operational since 1972, has accumulated over 50 years of global land surface temperature data, forming a long-term series archive containing over 9 million images with continuously rapid growth [10].

1.2.2. Technological Development History and Accuracy Improvement

Urban heat island remote sensing monitoring technology has achieved leapfrog improvements in core technical indicators since its development from the late 1970s to the present. In terms of technical precision, spatial resolution has significantly improved from the kilometer level to tens of meters, with the highest reaching 70 m resolution [11], making fine-scale identification of urban internal thermal environments possible. Temperature measurement accuracy has continuously improved, with operational algorithms achieving root mean square errors of 2.2–2.3 K [12] and new algorithms controlling errors within 1.5 K in simulation tests [13], significantly enhancing the reliability of heat island intensity calculations. The temporal resolution has also evolved from traditional 16-day revisit cycles to minute-level high-frequency observations [14], providing strong technical support for capturing dynamic changes in heat islands.

Accompanying the enhancement of technical capabilities, the application scale of heat island remote sensing research has shown rapid expansion. Satellite-based heat island research has grown exponentially, with the number of studies published during 2006–2010 (74 papers) showing a 40% increase compared to the total from 1972–2005 (53 papers) and annual publications in recent years [15], fully demonstrating the synchronized improvement of technical maturity and application value. The specific implementation pathways and representative platform characteristics of these technological advances will be analyzed in detail in Chapter 3 on Remote Sensing Technology Evolution.

1.2.3. Core Application Values of Remote Sensing Technology

The key roles of remote sensing technology in heat island research primarily manifest in five aspects: First, through thermal infrared data combined with methods such as Mono-Window Algorithm (MWA) and Split-Window Algorithm (SWA), it enables the precise retrieval of land surface temperature across large areas and multiple scales; second, it captures the dynamic temporal changes and long-term evolutionary trends of heat islands using multi-temporal data; third, through the combined analysis of thermal infrared and multispectral data, it reveals the quantitative relationships between land cover types, urban density, green space distribution, and thermal environments; fourth, compared to large-scale ground monitoring networks, it provides a more cost-effective solution for heat island observation. Research demonstrates that satellite remote sensing monitoring can save USD millions to hundreds of millions in annual costs compared to in situ measurements [16]; fifth, it establishes a scientific foundation for formulating and evaluating heat island mitigation strategies.

1.2.4. Inherent Limitations and Technical Challenges

Although remote sensing technology demonstrates significant advantages in urban heat island research, it still faces multiple inherent limitations that affect the accuracy and reliability of UHI assessment, requiring researchers to give full consideration to applications.

Accuracy and precision limitations constitute the primary challenge in remote sensing UHI monitoring. Hu and Brunsell [17] found through validation studies of four major North American cities that near-surface air temperatures obtained from MODIS atmo-

spheric profile products had accuracies of 3–7 K RMSE compared to ground observations across different cities and time periods, with extremely dry (Phoenix) and extremely humid (Houston) climate conditions increasing the variability of MODIS temperature accuracy. Additionally, MODIS air temperatures and dew point temperatures were generally underestimated [17], with this systematic bias showing inconsistent performance across different time periods and regions.

Data acquisition and environmental factor limitations severely affect the availability and quality of remote sensing data. Degefu et al. [18] pointed out that satellite-acquired LST data are severely affected by cloud cover and other factors, such as viewing angle effects and sensor internal errors. A research case showed that among 21 images viewed in the United States, only one image was cloud-free [18]. Urban areas have a higher probability of afternoon cloud formation due to strong convective activities, further limiting data acquisition frequency. Meanwhile, Du et al. [19] studied over 5500 cities globally and demonstrated that urban thermal anisotropy effects are significant. When using large viewing angle ($\pm 60^\circ$) satellite data, urban surface sensible heat flux and heat island intensity can be underestimated by 45.4% and 43.0%, respectively, with such viewing angle effects being widespread across global cities.

The inherent trade-off between spatial and temporal resolution constitutes the core technical bottleneck in remote sensing UHI monitoring. Degefu et al. [18] pointed out that Landsat has two disadvantages: satellite revisit time (16 days) and image size, making the images unsuitable for monitoring UHI effect changes within a day or week. While MODIS records images at spatial resolutions of 250 m (bands 1–2), 500 m (bands 3–7), and 1 km (bands 8–36), thermal infrared images are captured at 1 km resolution, and due to spatial resolution limitations, these images are mainly used for large study area research [18]. This spatio-temporal resolution trade-off makes it difficult for researchers to simultaneously obtain high-precision spatial details and sufficient temporal dynamic information.

The fundamental differences between land surface temperature and air temperature present conceptual challenges for remote sensing UHI research. Hu and Brunsell [17] pointed out that while satellite-observed LST can provide a way to estimate air temperature distributions, the relationship between LST and air temperature remains empirical. To link LST and air temperature, many factors must be considered, such as surface properties, atmospheric conditions, and solar angles [17]. The authors emphasized that air temperature directly retrieved from satellite remote sensing may have greater application potential because it avoids the scaling issues and problems associated with different essential physical factors impacting these two types of temperatures [17]. These fundamental differences limit the direct application of remote sensing results in human comfort and public health assessments.

Methodological standardization difficulties constrain the comparability and application value of UHI research results. Degefu et al. [18] pointed out that traditional UHI intensity quantification methods using paired measurements are very sensitive to site or grid selection, and the highly diverse UHI estimates used by different studies make cross-city or cross-study comparisons extremely challenging. Cases cited in the research show that when Degefu et al. applied 11 UHI assessment techniques to 263 European cities, the results showed weak negative correlations between estimated UHIs [18], further highlighting the urgent need for methodological standardization.

Validation and accuracy assessment challenges limit the accurate evaluation of remote sensing UHI research reliability. Degefu et al. [18] emphasized that major obstacles relate to adjustments for radiative error consequences and land surface emissivity, with existing validation methods having limitations. Particularly, many studies using Landsat and MODIS analyzed in the review did not report the accuracy of retrieved LST [18]. Ground measure-

ment networks often cannot match satellite observation coverage in spatial distribution, and factors such as temporal differences between satellite overpass times and ground observations all increase the complexity of validation work. Based on a deep understanding of these limitations, this research is committed to overcoming these challenges through technological and methodological innovations.

These inherent limitations do not negate the important value of remote sensing technology in UHI research but rather remind researchers of the need to fully recognize and reasonably address these limitations in applications. Through multi-source data fusion, algorithm improvements, validation strategy optimization, and methodological standardization, the impact of these limitations can be effectively mitigated, improving the accuracy and reliability of remote sensing UHI monitoring.

1.3. Research Questions and Review Objectives

This study adopts a systematic narrative review methodology, drawing upon the systematic principles and reporting standards of the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) guidelines. While structured frameworks such as PICO also have important application value in environmental research, considering the narrative characteristics and multi-dimensional analysis requirements of this review, the PRISMA framework is more suitable for the systematic literature review methodology of this study, ensuring transparency, reproducibility, and methodological rigor of the research process.

Despite urban warming emergence being a globally concerning environmental issue, and with satellite-based observation systems making a significant contribution to heat island monitoring, current research still faces multiple challenges. The primary technical method integration issue involves how to effectively integrate evolving remote sensing technologies, data processing methods, and urban thermal environment analysis models to establish a more comprehensive and precise urban heat island monitoring system. Related to this is the scale matching problem: how to resolve the mismatch between the spatial-temporal resolution trade-off of remote sensing observations and the multi-scale characteristics of urban heat islands to achieve continuous monitoring from urban regions to neighborhood microclimates. Meanwhile, the data validation problem in complex urban environments constrains the reliability of remote sensing heat island research, necessitating more effective validation strategies and accuracy assessment systems. From an application perspective, how to transform remote sensing monitoring results into practical tools supporting urban planning decisions and heat island mitigation, especially enhancing urban thermal resilience in the context of climate change, is also an urgent problem to be solved.

Based on the above challenge analysis, this review explicitly proposes four inter-related core research questions to construct a systematic knowledge framework for urban heat island remote sensing monitoring and assessment:

Research Question 1 (RQ1): How can we effectively integrate continuously developing remote sensing technologies with data processing methods to enhance the technical capabilities of urban heat island monitoring? This question aims to systematically review the technological evolution from early traditional remote sensing platforms (Landsat, NOAA-AVHRR, and DMSP-OLS) to new-generation satellite systems (Sentinel series, ECOSTRESS, GOES-R, etc.), analyze the value of the contributions of spatial and temporal resolution improvements and multi-source data fusion technology development in urban heat island research, and evaluate the performance differences, advantages, and limitations of different remote sensing platforms in heat island monitoring.

Research Question 2 (RQ2): How can we resolve the mismatch between the spatial-temporal resolution trade-off of remote sensing observations and the multi-scale charac-

teristics of urban heat islands? This question is dedicated to comprehensively evaluating the progress of land surface temperature extraction algorithms (Mono-Window Algorithm MWA, Split-Window Algorithm SWA, and urban-specific algorithms) and heat island intensity calculation methods, deeply exploring the innovative development of multi-source remote sensing data integration and fusion technologies (spatiotemporal fusion, downscaling techniques, machine learning methods, multi-sensor integration, and physical model integration) and analyzing how to overcome the limitations of single data sources through technological innovation.

Research Question 3 (RQ3): How can we establish more effective validation strategies and accuracy assessment systems in complex urban environments? Based on the high complexity and heterogeneity of urban environments, this question systematically constructs a comprehensive accuracy assessment framework, including ground truth validation, cross-platform validation, and temporal validation, combined with statistical indicators, spatial analysis, and temperature comparison methods to ensure the scientific reliability of remote sensing heat island research results and provide a methodological foundation for comparability between different studies.

Research Question 4 (RQ4): How can we transform remote sensing monitoring results into practical tools supporting urban planning decisions and heat island mitigation? This question focuses on the practical application value of remote sensing technology, revealing the scientific patterns of the relationships between urban scale, morphology, and heat islands through typical case comparisons, analyzing the characteristic differences and commonalities of different types of urban heat islands, evaluating the application effects of remote sensing technology in heat island mitigation practices such as heat vulnerability area identification, green space planning, and urban material optimization, and exploring remote sensing-based heat island mitigation decision support systems.

To achieve the above research objectives, this study will systematically review the evolution of urban heat island remote sensing monitoring technology from early traditional platforms to the latest satellite systems, analyzing the value of technological progress for urban heat island research; comprehensively evaluate the methodological advances in land surface temperature extraction and heat island intensity calculation, comparing the applicable conditions, accuracy characteristics, and limitations of different algorithms; deeply explore the innovative development of multi-source remote sensing data integration and fusion technologies, analyzing how to overcome the limitations of single data sources through data fusion; reveal the scientific patterns of the relationships between urban scale, morphology, and heat islands through typical case comparisons, analyzing the characteristic differences and commonalities of the different types of urban heat islands; evaluate the application value of remote sensing technology in heat island mitigation practices, including decision support roles in heat vulnerability area identification, green space planning, and urban material optimization; analyze the limitations of existing technologies and methods, and explore the application prospects of artificial intelligence and other emerging technologies in urban heat island monitoring, as well as the development trends in urban heat island monitoring under the background of climate change.

To ensure the scientific rigor and systematicity of literature selection, this study establishes clear inclusion and exclusion criteria systems. The inclusion criteria require the literature to simultaneously meet the following conditions: the research content involves remote sensing monitoring, assessment, or the application of urban heat island effects; the use of satellite remote sensing, thermal infrared remote sensing, or related technologies for heat island research; proposing new monitoring methods, algorithm improvements, or application strategies; research based on actual remote sensing data analysis with clear study areas and time ranges; and publication in peer-reviewed journals or important inter-

national conferences with clear methodological descriptions. The exclusion criteria include studies primarily focusing on atmospheric urban heat islands rather than surface heat islands; studies based solely on ground observations or numerical simulations without involving remote sensing technology; studies with highly repetitive content published by the same authors; and conference papers with only abstracts or low-quality regional conference papers, etc. Literature screening adopted a multi-stage progressive process through database retrieval, deduplication processing, title and abstract screening, full-text evaluation, and other steps, combined with snowball sampling methods to supplement the important literature, ultimately determining high-quality literature for inclusion in analysis through systematic screening. The specific literature retrieval strategies, search term combinations, database selection, and quality assessment methods are detailed in Section 2 Materials and Methods.

The theoretical contribution of this review lies in systematically integrating the technical and methodological systems of urban heat island remote sensing monitoring, identifying key challenges in research, and foreseeing future development directions; the practical value is reflected in providing scientific evidence for urban planners and decision-makers, supporting the formulation and evaluation of urban heat island mitigation strategies, and ultimately serving urban sustainable development and climate change adaptation. Figure 1 presents the comprehensive methodological framework for urban heat island remote sensing monitoring and assessment constructed in this study, systematically showing the complete research process from remote sensing data acquisition, land surface temperature extraction, heat island intensity calculation, and characteristic analysis in terms of heat island mitigation applications, as well as key technical methods, challenges, and typical findings at each stage. This framework not only summarizes the main content structure of this review but also reflects the technical roadmap and logical relationships of urban heat island remote sensing research. Through systematically answering the above research questions, this review strives to construct a comprehensive knowledge framework for the field of urban heat island remote sensing monitoring and assessment, providing a scientific basis for researchers in technical method selection, practical guidance for urban planners in heat island mitigation strategy formulation, and decision support for policymakers in urban climate adaptation planning, promoting the transformation of urban heat island remote sensing research from technological and methodological innovation to practical application.

To ensure the clarity and systematicity of the logical framework of this review, Table 1 presents the clear correspondence between the four core research questions and the content of each chapter, as well as the specific contributions of each chapter to answering the corresponding research questions. This systematic mapping relationship provides readers with a clear reading roadmap, ensuring that each research question receives comprehensive and systematic answers while demonstrating the complete logical development trajectory of this review from technological evolution to methodological innovation and then to practical applications.

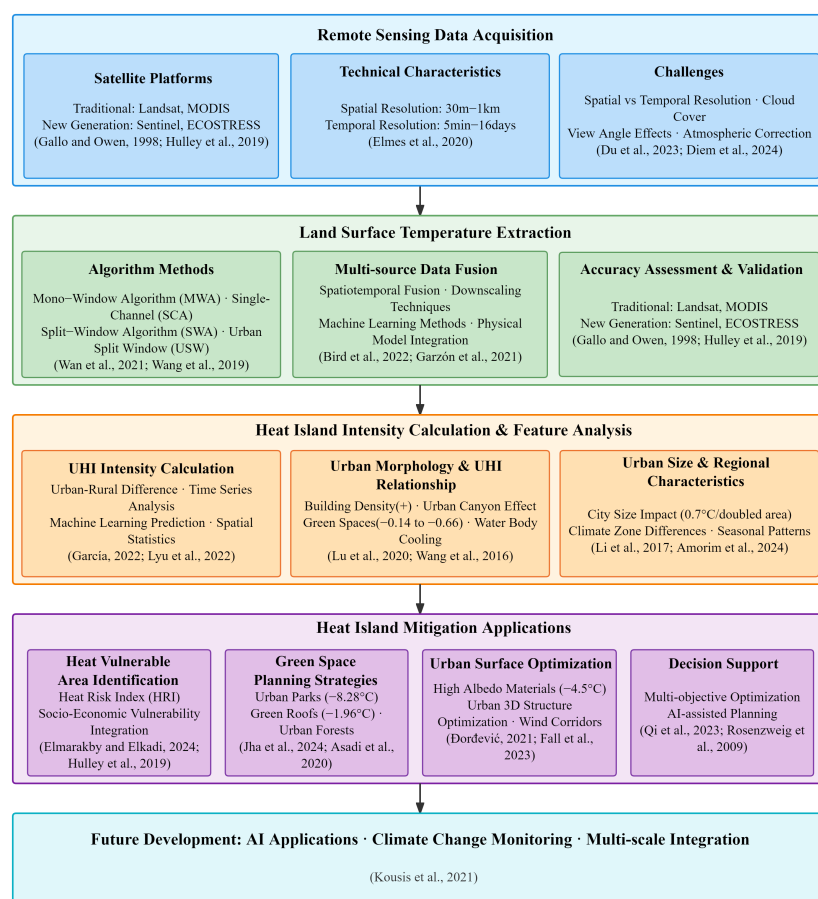


Figure 1. Comprehensive framework of remote sensing methods for urban heat island research [2,3,6,19–38].

Table 1. Systematic mapping between research questions and review chapters.

Research Questions	Primary Chapters	Supporting Chapters	Specific Contributions	Key Findings/Results
RQ1: How can we effectively integrate evolving remote sensing technologies and data processing methods to enhance urban heat island monitoring capabilities?	Section 3: Remote Sensing Technology Evolution	Section 1.2, Section 7.2	- Systematically trace technological evolution from 1972 Landsat-1 to next-generation systems like ECOSTRESS - Analyze spatial resolution improvement from traditional 1 km to 70 m - Evaluate temporal resolution enhancement from 16-day to GOES-R series 5 min monitoring - Construct multi-scale heat island monitoring spatial resolution adaptation framework	- Single Landsat scene covers 185×180 km, containing over 37 million 30 m pixel observation points - Landsat 8 TIRS achieves 0.4 K NEAT with LST algorithm RMSE below 1.5 K [13] - A 12-bit radiometric resolution provides 4096 gray levels, with 16-fold precision improvement [39]
RQ2: How can we address the mismatch between spatial-temporal resolution trade-offs in the remote sensing observations and multi-scale characteristics of urban heat islands?	Section 4: Methodological Advances	Section 3.2, Section 3.4	- Evaluate accuracy improvements of land surface temperature retrieval algorithms - Analyze multi-source data fusion techniques, including spatiotemporal fusion, downscaling, and machine learning methods - Construct systematic framework for ground truth validation, cross-platform validation, and temporal validation	- Mono-window algorithm achieves $RMSE < 2.4$ K and $R^2 > 0.9$ under optimal conditions [40] - Split-window algorithm RMSE ranges 0.51–1.8 K [41] - Urban-specific algorithms generally achieve RMSE below 1.0 K [42]

Table 1. Cont.

Research Questions	Primary Chapters	Supporting Chapters	Specific Contributions	Key Findings/Results
RQ3: How can we establish more effective validation strategies and accuracy assessment systems in complex urban environments?	Section 4.3: Accuracy Assessment and Validation Strategies	Section 4.1, Section 4.2	- Establish three major validation methods: ground truth validation, cross-platform validation, and temporal validation - Integrate three assessment techniques: statistical indicators, spatial analysis, and temperature comparison - Address validation challenges posed by high urban heterogeneity and complex environments	- Areas with tree canopy coverage > 47%: mean absolute error < 3.74 °C, $r^2 > 0.85$ [20] - Diversified validation methodology ensures result reliability - Urban environment adaptive validation techniques
RQ4: How can we transform remote sensing monitoring results into practical tools supporting urban planning decisions and heat island mitigation?	Section 6: Heat Island Mitigation Applications	Section 5: Heat Island Characteristic Comparative Studies	- Develop thermal vulnerability identification methods and green space planning strategies - Quantify cooling effects of different types of green spaces and high-albedo materials - Construct AI-driven decision support systems for heat island mitigation	- Urban parks achieve cooling effects up to 8.28 °C [21] - AI optimization strategies: temperature reduction 0.7–0.9 °C, cost savings 22.2–42.2% [6]

Note: RQ = research question; AI = artificial intelligence; RMSE = root mean square error.

1.4. Review Conceptual Framework

This review constructs a systematic conceptual framework for urban heat island remote sensing monitoring and assessment (Figure 2), with “Technology-Method-Application” as the core logical thread, reflecting the progressive relationship and mutual promotion among the three components.

At the technological development level, Section 3 systematically reviews the technological evolution from traditional remote sensing platforms to new-generation satellite systems, which provides a continuously improving observational capability foundation for heat island monitoring. Technological progress directly drives comprehensive improvements in spatial resolution, temporal resolution, and spectral capabilities, laying a solid foundation for subsequent methodological innovation.

Based on technological development achievements, Section 4 focuses on the methodological progress level, deeply analyzing the continuous improvement process of land surface temperature extraction algorithms, multi-source data fusion technologies, and accuracy verification methods. These methodological innovations effectively overcome technical limitations, forming a systematic methodological framework and significantly enhancing the accuracy and reliability of heat island monitoring.

Sections 5 and 6 embody the value transformation at the application practice level, demonstrating how to rely on advanced technologies and mature methods to widely apply heat island monitoring results to areas such as heat vulnerability identification, green space planning, and material optimization, generating significant socio-economic benefits. More importantly, the problems and needs discovered in application practice, in turn, drive technological innovation and methodological improvement, forming a virtuous cycle mechanism of “Technology-Method-Application-Feedback.” This conceptual framework not only reflects the development logic of urban heat island remote sensing research but also provides clear guidance for future research directions.

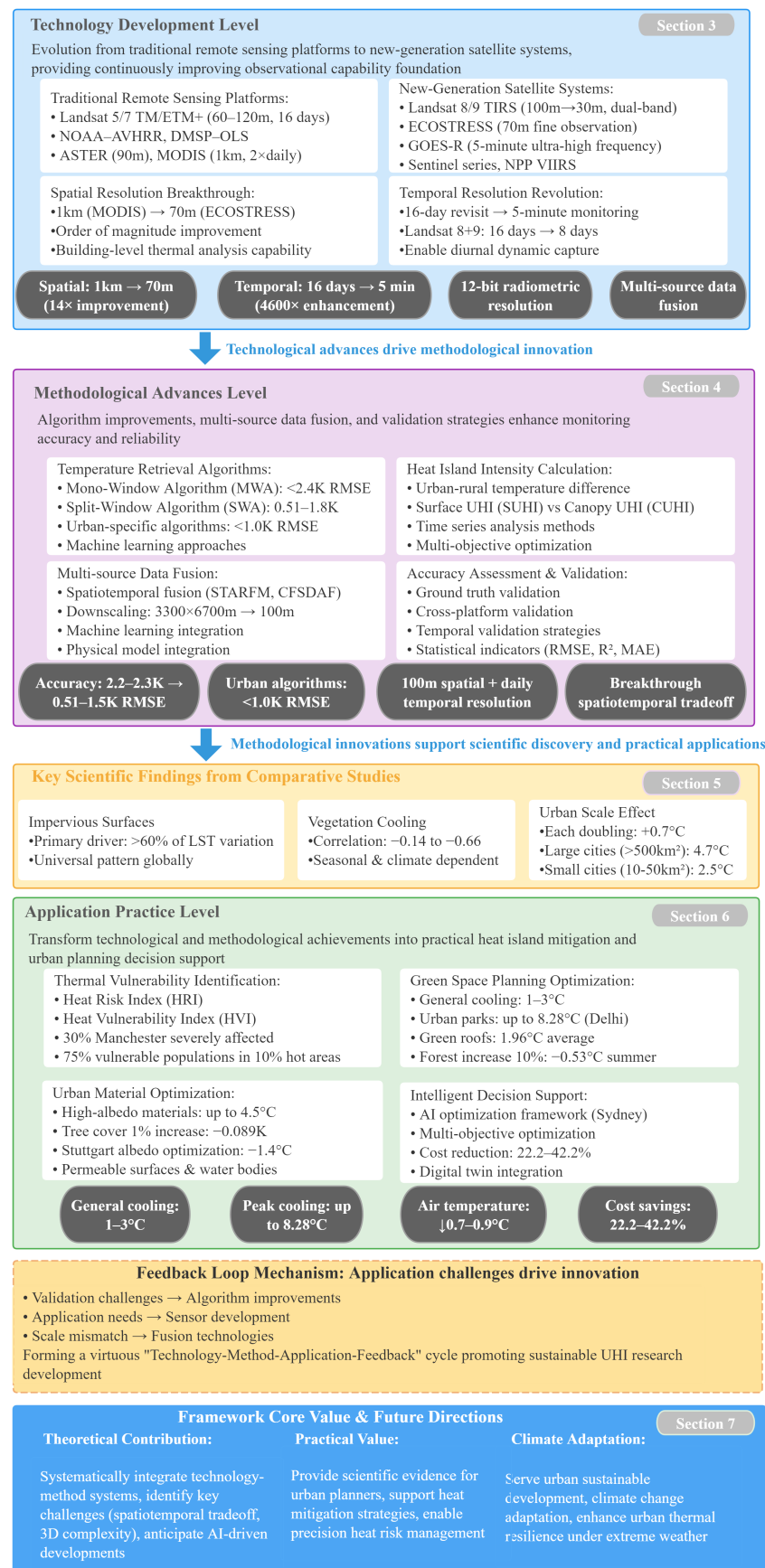


Figure 2. Urban heat island remote sensing monitoring and assessment conceptual framework: “Technology-Method-Application” progressive relationship and feedback loop.

2. Materials and Methods

2.1. Literature Search Strategy

This review employs a systematic literature search method to ensure comprehensive coverage of research related to remote sensing monitoring of urban heat island effects. The literature search was completed in December 2024, covering research literature from 1998 to December 2024. Major academic databases, including Web of Science Core Collection, Scopus database, IEEE Xplore Digital Library, ScienceDirect database, and Google Scholar, were searched to obtain the most comprehensive literature coverage.

The search strategy employed Boolean logic combinations of core keywords, specifically including subject terms: (“urban heat island” OR “UHI” OR “surface urban heat island” OR “SUHI” OR “canopy urban heat island” OR “urban temperature”) AND (“remote sensing” OR “satellite” OR “Landsat” OR “MODIS” OR “thermal infrared” OR “LST” OR “land surface temperature” OR “thermal remote sensing”) AND (“monitoring” OR “assessment” OR “detection” OR “mapping” OR “analysis” OR “evaluation”). The search time range was set from January 1998 to December 2024 with a language restriction so as to select only the English literature, as well as the following literature types: peer-reviewed journal articles, high-quality international conference papers, and authoritative technical reports.

2.2. Literature Screening Criteria and Process

Literature screening employed clear inclusion and exclusion criteria. The inclusion criteria required the literature to simultaneously meet the following conditions: research content involving remote sensing monitoring, assessment, or application in relation to urban heat island effects; the use of satellite remote sensing, thermal infrared remote sensing, or related technologies for heat island research; the proposal of new monitoring methods, algorithm improvements, or application strategies; research based on actual remote sensing data analysis with clear study areas and time ranges; and publication in peer-reviewed journals or important international conferences with clear methodological descriptions. The exclusion criteria included studies primarily focusing on atmospheric urban heat islands rather than surface heat islands; studies based solely on ground observations or numerical simulations without involving remote sensing technology; studies with highly repetitive content published by the same authors; conference papers with only abstracts or low-quality regional conference papers; literature with no access to the full text or incomplete key information; and non-English literature (unless of special importance).

Literature screening was conducted using a staged approach. Through database searches, 1134 relevant pieces of the literature were obtained, with 964 remaining after deduplication processing. Based on title relevance, preliminary screening was conducted, retaining 456 pieces of the literature. Subsequently, abstract screening was performed, reading abstracts and evaluating the relevance of research topics, methods, and contributions, retaining 198 pieces of the literature for full-text evaluation. In the full-text evaluation, the obtained full texts were strictly evaluated according to the inclusion and exclusion criteria, ultimately including 96 pieces of the literature. Additionally, snowball sampling was conducted through the references of the included literature to continuously supplement the important literature.

2.3. Data Extraction and Analysis Framework

Key information was extracted from each included literature piece, including basic research information (authors, year, journal, and study area), remote sensing data sources (satellite platforms, sensor types, and spatial/temporal resolution), technical methods (temperature retrieval algorithms, heat island calculation methods, and data fusion technologies), main findings (heat island intensity, influencing factors, and application effects), and research limitations and future directions. Quality assessment was conducted on the included litera-

ture, focusing on dimensions such as the rationality of research design, the scientificity of data processing methods, the adequacy of result validation, and the reliability and applicability of the conclusions, ensuring the review is based on high-quality research foundations.

The included literature consists primarily of journal articles, supplemented by important international conference papers and authoritative technical reports. The literature was mainly published in high-impact journals such as *Remote Sensing of Environment*, *Remote Sensing*, *ISPRS Journal of Photogrammetry and Remote Sensing*, and *Urban Climate*, as well as important international conferences in the remote sensing field such as IGARSS and International Symposium on Remote Sensing of Environment, ensuring the academic quality of the review. The publication years of the included literature span from 1998 to 2024, with a large proportion of the literature published after 2020, reflecting the rapid development of research in this field in recent years. Through the above systematic screening process, the high quality and relevance of the included literature are ensured.

The analysis framework was constructed according to preset research questions, including five main aspects: technological evolution (the development history of remote sensing platforms and sensors), methodological progress (land surface temperature extraction and heat island calculation methods), characteristic patterns (a comparison of different types of urban heat island characteristics), application practices (remote sensing support applications for heat island mitigation), and development trends (technical challenges and future development directions). It should be noted that this review has certain limitations, including language bias (mainly searching the English literature, potentially missing important research in other languages), publication bias (tendency to include studies with positive results), temporal limitations (publication up to December 2024), and uneven geographical coverage (the literature being mainly concentrated in developed countries and regions). Through the above systematic methods, this review strives to comprehensively and objectively present the current status and development trends of urban heat island remote sensing monitoring research, providing a reliable knowledge foundation for related research and applications.

It should be noted that, given the substantial variations in research methodologies, spatial scales, temporal coverage, and geographical contexts among the included studies, this review employs a qualitative synthesis approach for evaluating the significance of different factors rather than relying on simple frequency-based statistics. This methodological approach better captures the complexity of factor interactions, regional specificities, and inter-study methodological variations while mitigating the potential biases inherent in statistical inference procedures. Nevertheless, the analytical outcomes may still be subject to influences from literature coverage limitations and the authors' subjective interpretations, and variations in methodological rigor and regional representativeness across studies may impact the comprehensiveness of the synthesis.

3. Evolution of Remote Sensing Technology

The evolution of remote sensing technology is the technical foundation for the development of urban heat island monitoring. This chapter systematically reviews the technological development process from traditional platforms to new-generation systems, analyzing how technological progress lays the foundation for subsequent methodological innovation and application expansion. Specifically, this chapter will deeply analyze the implementation pathways of technological breakthroughs mentioned in Section 1.2, such as spatial resolution improvement (from 1 km to 70 m) and temperature measurement accuracy enhancement (reaching 0.1 °C), as well as how these technological innovations gradually overcome the inherent limitations described in Section 1.2, such as spatiotemporal resolution trade-offs and cloud coverage impacts.

Urban heat island remote sensing monitoring technology began in the late 1970s, developing and expanding alongside Earth observation satellite initiatives. The timeline in Figure 3 presents a panoramic view of the evolution in this field, from the first launch of the Landsat 1 satellite in 1972 to the widespread application of emerging technologies today, outlining key milestones in the development of remote sensing platforms and technologies. This trajectory of technological advancement has not only expanded the breadth and precision of heat island observations but also continuously promoted innovation in urban climate research methodologies.

Urban heat island remote sensing research in its early stages primarily relied on several pioneering remote sensing platforms. Although these platforms had many technical limitations by today's standards, they laid a solid scientific foundation for heat island research. The most representative among these were the Landsat series satellites, NOAA-AVHRR (Advanced Very High-Resolution Radiometer of the National Oceanic and Atmospheric Administration), and DMSP-OLS (Operational Linescan System of the Defense Meteorological Satellite Program). Each of these traditional platforms had its unique characteristics, establishing benchmarks in technical capabilities and application value for subsequent research. The following will deeply analyze the technical features of these pioneering platforms and their historical contributions to scientific research on urban heat islands.

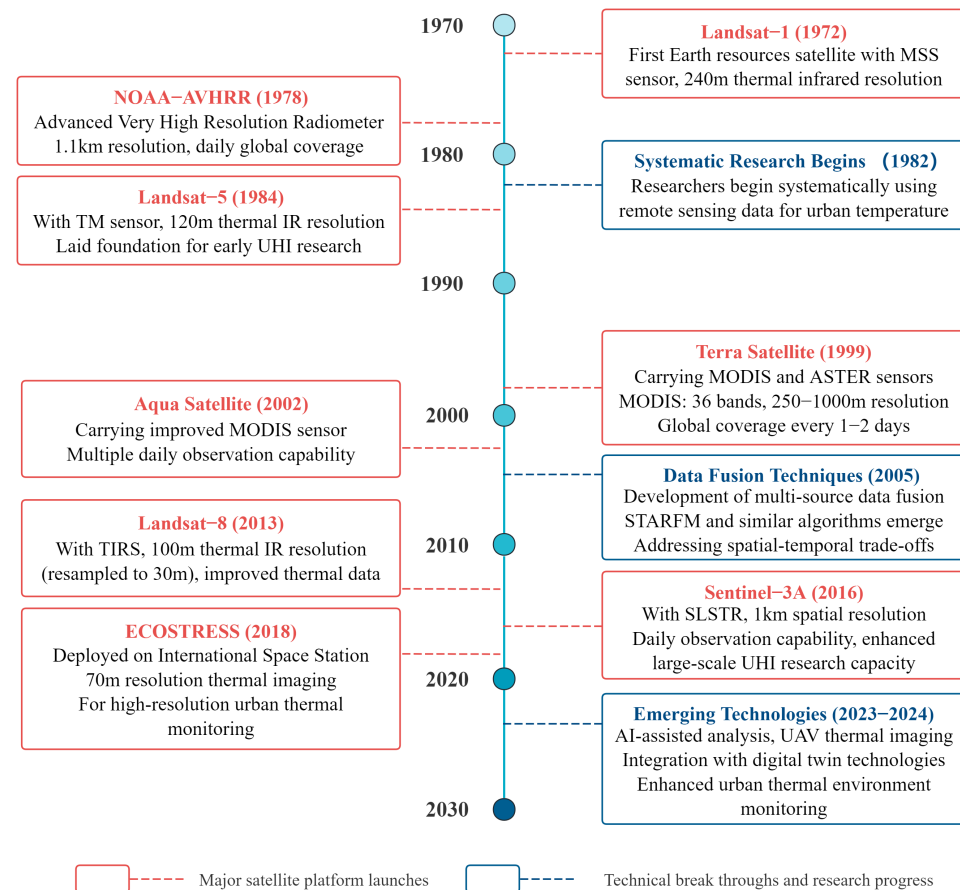


Figure 3. Evolution and application of remote sensing technologies for urban heat island monitoring.

3.1. Traditional Remote Sensing Platforms and Technical Characteristics

The Landsat series, as a pioneering satellite platform for urban heat island research, laid the technical cornerstone for heat island monitoring from the Thematic Mapper (TM) on Landsat-5 to the Enhanced Thematic Mapper Plus (ETM+) on Landsat-7. The 60–120 m thermal infrared spatial resolution provided by these sensors made medium-precision mapping of urban temperatures possible. Table 2 shows that although Landsat 5/7 had

advantages in capturing spatial details, their 16-day revisit cycle made it difficult to capture short-term dynamic changes in urban thermal environments.

Table 2. Comparison of major remote sensing platforms in urban heat island research.

Platform	Launch/ Operation	Spatial Resolution	Temporal Resolu- tion	Thermal IR Bands	Advantages	Limitations	Applications
Traditional Remote Sensing Platforms							
Landsat 5/7	1984– 2013/1999– present	60–120 m	16 days	1 band	Higher spatial res., long-term continuity	Low temporal res., cloud impact	Gallo and Owen [22] multi-sensor compari- son
NOAA- AVHRR	1978–present	1.1 km	Daily	2 bands	High tempo- ral res., global coverage	Low spatial res., mixed pixels	Huang et al. [43] global UHI climatology
ASTER	1999–present	90 m	16 days	5 bands	High spec- tral/spatial res.	Non-routine obs., limited coverage	Zhou et al. [15] SUHI platform review
MODIS	1999/2002– present	1 km	1–2/day	16 bands	High-freq. obs., global coverage	Low spatial res.	Li et al. [44] LST series (1985–2019)
New Generation Satellite Systems							
Landsat 8/9	2013/2021– present	100 m (to 30 m)	16 days	2 bands	Improved radio- metric res. (12- bit)	Limited temporal res.	Islam et al. [45] high- precision UHI mapping
Sentinel-2	2015/2017– present	No thermal IR (10–20 m)	5 days	None	High spatial res., land cover class.	Cannot measure LST	Piestova et al. [46] ther- mal domains
Sentinel-3	2016/2018– present	1 km	Daily	Multiple	High tempo- ral res.	Low spatial res.	Sobrino and Irakulis [47] global UHI
ECOSTRESS	2018–present	70 m	3–5 days	5 bands	Fine spa- tial/spectral res.	Non-systematic coverage	Hulley et al. [23] fine- scale mapping
GOES-R	2016–present	2 km	5–15 min	Multiple	Ultra-high tempo- ral res.	Low spatial res.	Bah et al. [48] spatial downscaling to 30 m
NPP VIIRS	2011–present	375–750 m	Daily	Medium res.	Enhanced night- time imaging	Mixed pixel issues	Khan et al. [49] UHI and urban expansion
Emerging Data Fusion Technologies							
Multi-source fusion	-	30–100 m	Daily+	Combined	High spa- tial/temporal res.	Complex process- ing	Bird et al. [24] 100 m daily LST
Spatiotemporal downscaling	-	30–100 m	Daily+	Improved low-res	Enhanced resolu- tion	Data quality de- pendent	Garzón et al. [25] data integration
ML fusion	-	Variable	Variable	Intelligent	Nonlinear proc., precision	Large training datasets	Yang and Lee [50] scale separation

Note: IR = infrared, LST = land surface temperature, UHI = urban heat island, ML = machine learning, res. = resolution.

ASTER (Advanced Spaceborne Thermal Emission and Reflection Radiometer), as another important traditional sensor mounted on the Terra satellite, provided a more refined observational perspective for urban microclimate research with its 90 m thermal infrared resolution. Zhou et al. [15] comprehensively reviewed satellite remote sensing applications for surface urban heat island analysis, highlighting how ASTER, Landsat, and MODIS data integration provides complementary perspectives for understanding urban thermal environments across different spatial and temporal scales. MODIS (Moderate Resolution Imaging Spectroradiometer), with its capability for global coverage twice daily, compensated for the temporal frequency limitations of high-resolution sensors, although its 1 km spatial resolution restricted applications to regional-scale heat island research. The global canopy urban heat island climatology atlas created by Huang et al. [43] using MODIS data

fully demonstrated the unique advantages of this sensor in identifying large-scale heat island patterns.

The fundamental challenges faced by traditional remote sensing technology in heat island monitoring are manifested in four aspects: the inherent trade-off between spatial and temporal resolution, the complexity of atmospheric correction, signal ambiguity caused by mixed pixels in urban environments, and the limited night-time observation capabilities of early platforms.

3.2. New Generation Satellite Systems and Their Advantages

Since the 21st century, new-generation satellite systems have completely revolutionized urban heat island monitoring capabilities. As shown in Table 2, Landsat 8/9 and the Sentinel series satellites lead this technological leap, opening a new era for fine-scale urban heat island research. Compared to traditional systems, new-generation satellites have achieved comprehensive upgrades in key technical indicators such as spatial precision, temporal frequency, spectral capabilities, and radiometric performance, enabling urban heat island research to evolve from simple static observations to a multi-dimensional, high-precision comprehensive analysis of urban thermal environments.

Leapfrog improvements in spatial resolution have laid the foundation for fine-scale urban heat island monitoring. The Thermal Infrared Sensor (TIRS) aboard Landsat 8 (2013) and Landsat 9 (2021) provides a 100-m thermal infrared resolution (resampled to 30 m) [51,52], achieving qualitative improvements over the earlier 60–120 m resolution. This improvement brings significant monitoring advantages: first, fine spatial resolution effectively reduces mixed pixel problems in urban environments, improving the accuracy of temperature extraction and being particularly suitable for thermal environment analysis in medium-density urban areas; second, the new resolution can precisely capture spatial variations in urban internal thermal environments, identifying temperature differences and hotspot distributions at the neighborhood scale. ECOSTRESS represents an important breakthrough in spatial thermal imaging, with its 70-m resolution achieving an order-of-magnitude improvement compared to the 1-km resolution [11] of MODIS, enabling a clear distinction regarding the thermal characteristics of individual urban elements such as buildings, parking lots, and small green spaces; this allowed researchers to conduct building-level thermal environment analysis at a satellite scale for the first time. Hulley et al. [23] achieved unprecedented urban fine-scale temperature mapping using ECOSTRESS, revealing rich spatial details of thermal environments.

Significant improvements in temporal resolution provide crucial support for understanding heat island dynamic processes. Traditional 16-day revisit cycles make it difficult to capture rapid changes in urban thermal environments, while the temporal performance improvements of new-generation systems make several key applications possible. The GOES-R series, with its ultra-high temporal frequency of 5 min [48], makes the analysis of heat island diurnal dynamic changes possible. Wang et al. [53] achieved high spatial resolution temperature monitoring at a 100-m resolution by downscaling MODIS data using random forest models, providing enhanced data sources for surface urban heat island analysis. During extreme weather events, high-frequency observations provide real-time monitoring of heat island enhancement effects during heat waves, with García [26] confirming the important value of such high temporal resolution observations in heat wave–heat island coupling studies. Additionally, multi-satellite constellations significantly improve data acquisition continuity, with Masek et al. [54] noting that the joint operation of Landsat 8 and 9 reduces the revisit cycle from 16 days to 8 days, providing a more stable data foundation for heat island long-term trend analysis.

Comprehensive improvements in spectral capabilities have created conditions for the precise identification of heat island driving factors. The Sentinel series of the European Copernicus program has contributed unique value to heat island research. Although Sentinel-2 does not have thermal infrared bands, its 10-m multispectral resolution and 13 spectral bands [55] can precisely distinguish different vegetation types, building materials, and surface features, providing high-quality land cover input data for thermal environment analysis. The urban thermal microclimate mapping conducted by Piestova et al. [46] combining Landsat and Sentinel-2 data demonstrates the application potential of this platform in fine-scale thermal environment analysis. The joint analysis of multispectral vegetation indices (such as NDVI, EVI, SAVI, etc.), combined with high spatial resolution land cover classification, enables a more accurate and reliable quantitative assessment of vegetation cooling effects and the thermal characteristics of urban surface materials. The Sea and Land Surface Temperature Radiometer (SLSTR) of Sentinel-3, which has daily observation frequency and a multi-thermal infrared band design, enhances temperature estimation accuracy. The global surface heat island analysis method constructed by Sobrino and Irakulis [47] using Sentinel-3A SLSTR data validates its significant advantages in large-scale heat island research.

Significant improvements in radiometric performance and measurement accuracy directly translate into scientific value for heat island research. The TIRS dual-band design significantly improves atmospheric correction capabilities and temperature estimation accuracy by compensating for atmospheric effects through the differences between two thermal infrared bands at 10.9 μm (Band 10) and 12.0 μm (Band 11) [56], substantially enhancing temperature retrieval accuracy under complex atmospheric conditions. The 12-bit radiometric resolution of Landsat 8/9 (compared to earlier 8-bit) greatly enhances the capability to capture subtle temperature changes. The 12-bit radiometric resolution represents a significant improvement over the earlier 8-bit version, enabling sensors to identify more subtle temperature variations and significantly enhancing temperature measurement precision, which is of great value for detecting minor changes and seasonal fluctuations in urban heat island intensity. The pushbroom scanning design, compared to traditional whisk-broom methods, significantly extends pixel integration time, improving Noise Equivalent Delta Temperature ($\text{NE}\Delta\text{T}$) performance and substantially enhancing the signal-to-noise ratio of thermal infrared data [39]. The cumulative effects of these accuracy improvements enable heat island monitoring to transition from qualitative description to precise quantitative analysis, providing more of a scientific basis for urban planning decisions. Islam et al. [45] demonstrated through comparative analysis that the improvements in temperature calculation efficiency and atmospheric correction capability of Landsat 9 effectively enhanced urban heat island identification and mapping accuracy compared to Landsat 8. Elmes et al. [20] found that in areas with tree canopy coverage exceeding 47%, Landsat 8 temperature estimation achieves high accuracy levels (mean absolute error $< 3.74\text{ }^{\circ}\text{C}$; $r^2 > 0.85$).

Other important satellite systems further enrich the technical means for heat island monitoring. NPP VIIRS, with its enhanced night-time imaging capabilities, fills critical gaps in night-time heat island research. Khan et al. [49] skillfully combined Landsat 8 and NPP VIIRS data to achieve a comprehensive analysis of heat islands and urban expansion. Chinese indigenous satellites, such as GaoFen-1 and HJ-1B, have also joined the heat island research community, with Liu et al. [57] fusing MODIS, Landsat 8, and GaoFen-1 data to create high-resolution temperature time series, demonstrating the application potential of Chinese satellite technology in heat island monitoring.

In summary, the technological advantages of new-generation satellite systems have achieved a comprehensive leap in urban heat island monitoring from “observational capa-

bilities” to “analytical depth.” The synergistic enhancement of spatial, temporal, spectral, and radiometric resolutions not only strengthens individual technical indicators but, more importantly, creates multi-dimensional, high-precision, comprehensive observation capabilities for urban thermal environments. Land surface temperature measurement accuracy has achieved significant improvements, with Landsat 5 and 7 thermal infrared sensors showing root mean square errors of 2.2–2.3 K in ground validation [12], while the dual-band design of Landsat 8 TIRS enables newly developed single-channel and split-window algorithms to achieve average errors below 1.5 K in simulation validation, with the split-window algorithm performing better at approximately 1 K RMSE [13]. Spatial resolution has leaped from 1 km to 70 m, and temporal resolution has shortened from 16 days to 3–5 days [23]. These technological breakthroughs make the reliable detection of subtle heat island intensity variations possible and enable the independent identification of thermal characteristics of individual urban elements, such as buildings, providing data support for building-level heat island mitigation strategies. This qualitative leap in observational capabilities establishes a solid technical foundation for the methodological innovations to be discussed in Section 4 and the application expansions in Sections 5 and 6, driving heat island research from traditional phenomenological description toward mechanistic understanding and precision management.

3.3. Improvements in Spatial and Temporal Resolution and Their Application Value

As shown in Table 2, key breakthroughs in remote sensing technology for urban heat island research lie in the continuous improvement of spatial and temporal resolution and innovation in data integration methods, which have greatly expanded the application value of heat island monitoring. Figure 4 intuitively displays the distribution pattern of major remote sensing platforms in the spatiotemporal resolution dimension, clearly presenting the evolutionary trajectory and complementary relationships from traditional platforms to new generation systems. As seen in the figure, traditional platforms (gray dots) and new generation platforms (orange dots) form distinctly different clusters, with new generation platforms achieving overall better spatiotemporal resolution combinations, while multi-source data fusion technologies strive to break this inherent trade-off between spatial and temporal resolution.

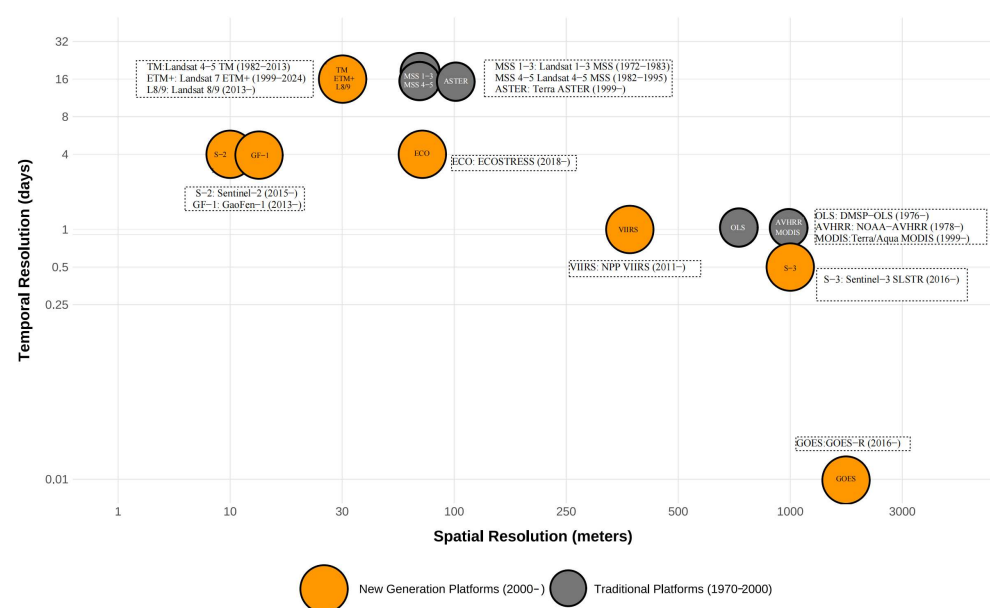


Figure 4. Comparison of spatiotemporal resolution of remote sensing platforms.

The improvement in spatial resolution has enabled heat island research to progress gradually from the macro-regional level to the micro-urban structure level. Developing from an early, coarse resolution of 1 km to modern sub-meter precision observation, this advancement allows researchers to precisely analyze the intrinsic relationship between urban structure and thermal environment. Liu et al. [58] applied a high-resolution Thermal Airborne Spectrographic Imager (TASI, 0.6/1.25 m) to analyze the thermal characteristics of urban materials, providing direct guidance for heat island mitigation strategies. High spatial resolution also supports fine-scale microclimate monitoring; Nichol's [59] research confirmed that increasing resolution from 90 m to 10 m significantly enhanced the ability to identify microscale temperature patterns, precisely locating hot spots and cool island effects within cities and providing a scientific basis for targeted cooling measures. Additionally, a comparison between high-resolution data and ground temperature observation stations indicates that improved sensor resolution directly enhances consistency between remote sensing and ground measurements.

The leap in temporal resolution from observations once every 16 days (Landsat series) to today's high-frequency monitoring every day or even every 5 min (GOES-R), as shown in Figure 4, has greatly expanded the dimension of heat island temporal dynamics research. High temporal resolution enables researchers to comprehensively capture the day-night variation patterns of heat islands, revealing the dynamic processes of their formation and dissipation. Wang et al. [1] used MODIS data with four times daily observations to analyze the differences between canopy and surface heat islands caused by air advection, a type of research that depends on high-frequency temperature observations. Long-term high-frequency observations also support studies on the seasonal variations and long-term trends of heat islands; Li et al. [44] used a long-term temperature series reconstructed from Landsat data from 1985–2019 to reveal the long-term dynamic evolution of heat islands in China's major cities. García [26] applied Sentinel-3 data to study heat island changes during heat waves, with high temporal resolution observations making it possible to study heat island responses under extreme weather conditions.

To overcome the inherent limitations of single sensors in spatiotemporal resolution, researchers have developed various innovative data integration methods. The emerging data fusion technologies shown in Table 2 demonstrate these breakthrough methods. Bird et al. [24] combined Landsat and MODIS data to simultaneously achieve temperature monitoring with a 100 m spatial resolution and a daily temporal resolution, demonstrating the application value of spatiotemporal fusion technology. Bechtel et al. [60] downsampled geostationary satellite data from 3300×6700 m to 100 m while maintaining a 15 min observation frequency, significantly enhancing spatial details for high-frequency heat island monitoring. Garzón et al. [25] applied machine learning techniques to integrate Landsat and Sentinel-2 data for heat island modeling, achieving high-precision prediction and analysis. Shi et al. [61] developed a Comprehensive Flexible Spatiotemporal Data Fusion (CFSDAF) method to generate high spatiotemporal resolution urban land surface temperature images, fully leveraging the complementary advantages of different satellite sensors while considering the highly heterogeneous urban surface characteristics. The temperature product fusion framework created by Yang and Lee [50] effectively integrated MODIS and Landsat 8 data, enhancing the spatiotemporal continuity of heat island research.

Improvements in spatiotemporal resolution and innovations in data integration have greatly expanded the application value of heat island research. High-resolution heat island data enable urban planners to implement precise heat island mitigation designs at the neighborhood scale. Seeberg et al. [62] used Landsat data to evaluate the impact of Stuttgart's greening policies on heat islands, providing a scientific basis for urban greening. High-precision heat island mapping makes it possible to identify thermally vulnerable areas

within cities, providing heat risk protection support for vulnerable groups. Long time series high-resolution data support research on the relationship between urban development and thermal environments; Nandi and Dede [63] analyzed heat island changes in the rapidly urbanizing areas of West Java from 1998–2018, providing a basis for sustainable urban development. Improved remote sensing data provide precise boundary conditions and validation data for microclimate numerical simulations, enhancing simulation reliability. The combination of high spatiotemporal resolution heat island data with health data supports research on the relationship between thermal environments and resident health, providing scientific support for public health policies.

Despite these advances, heat island remote sensing monitoring still faces an inherent trade-off between spatial and temporal resolution. As shown by the platform characteristics compared in Table 2, Liu and Weng [64] found that there are significant scale effects in the relationship between urban landscape patterns and surface temperature, with the optimal resolution for analyzing heat islands differing between being category-based (30 m) and at the landscape level (90 m). This finding indicates that selecting appropriate resolution parameters is crucial for the accuracy of heat island research results. With continuous innovation in satellite technology and data processing algorithms, heat island remote sensing monitoring capabilities are expected to improve further, providing a more comprehensive and precise scientific basis for urban sustainable development and climate adaptation planning.

3.4. Spatial Resolution Adaptation for Multi-Scale Heat Island Monitoring

The multi-scale characteristics of urban heat islands require remote sensing monitoring to adopt corresponding spatial resolutions and technical methods at different planning levels. Based on the spatial extent and planning requirements, urban heat island research can be divided into three main scale levels, each corresponding to different spatial resolution requirements, data source selections, and application objectives.

The Urban-Regional Scale primarily focuses on the heat island pattern analysis of entire cities or metropolitan areas, suitable for regional planning and policy formulation. This scale typically employs medium spatial resolution data (1 km level), such as MODIS LST products, which can effectively capture urban–rural thermal differences and large-scale heat island spatial distributions. Huang et al. [43] used MODIS data from 2000–2013 to create urban heat island climatology for Shanghai, demonstrating the advantages of this scale in long-term heat island dynamic studies. The global study of Zhang et al. [5] shows that cities with areas exceeding 500 square kilometers have average summer daytime heat island intensities of 4.7 °C, while cities with areas between 10–50 square kilometers average 2.5 °C, reflecting the relationship between urban scale and heat island intensity. Monitoring results at this scale primarily support urban master planning, heat island mitigation policy formulation, and regional sustainable development strategies.

The neighborhood scale focuses on thermal environment differences among different functional zones and communities within cities, providing support for refined urban planning. This scale requires high spatial resolution data (30–100 m), primarily using Landsat series as the main data source, capable of identifying the impacts of different land use types, building densities, and green space distributions on thermal environments. Liu and Weng's [64] study in Indianapolis shows that the optimal resolution for analyzing landscape-temperature relationships varies by research level: category-based analysis is suitable for a 30 m resolution, while landscape-level analysis is more appropriate for a 90 m resolution, providing important methodological guidance for community-scale research. Monitoring at this scale supports community heat vulnerability identification, green space planning and layout, and building environment optimization.

The microclimate scale targets fine-scale thermal environment analysis of building clusters, neighborhood interiors, or individual plots, serving building design and microenvironment improvement. This scale requires ultra-high spatial resolution data (<30 m), typically employing high-resolution satellite data or airborne sensors. Liu et al. [58] analyzed urban heat island effects in Shijiazhuang using Landsat TM (30 m/120 m) and airborne thermal infrared spectral imager TASI (0.6/1.25 m) data, finding that high-resolution data can reveal subtle changes in land surface temperature, with temperature differences reaching 7 °C (satellite data) or 10 °C (airborne data) at noon, demonstrating the application value of microscale monitoring. ECOSTRESS represents an important breakthrough in spatial thermal imaging, with its 70 m resolution achieving an order-of-magnitude improvement compared to the 1 km resolution of MODIS, enabling clear distinction of thermal characteristics of individual urban elements such as buildings, parking lots, and small green spaces and allowing researchers to conduct building-level thermal environment analysis at the satellite scale for the first time.

Cross-scale data fusion and scale conversion are important approaches to address single-scale limitations. Nichol [59] validated the effectiveness of emissivity modulation methods by comparing the spatial distributions of enhanced-resolution thermal images with measured temperatures, demonstrating the value of high-resolution data in microscale temperature pattern recognition. Bechtel et al. [60] downscaled low-resolution data from geostationary satellites (3300 × 6700 m) to 100 m while maintaining high temporal resolution (15 min), demonstrating the application potential of downscaling techniques in cross-scale data conversion. These cross-scale techniques provide technical support for constructing multi-level heat island monitoring systems, enabling different planning levels to formulate corresponding heat island mitigation strategies based on consistent scientific foundations. Table 3 systematically summarizes the correspondence between spatial resolution and detection capabilities and mitigation strategies in multi-scale urban heat island monitoring, providing direct guidance for selecting appropriate remote sensing data sources and technical methods for different planning levels.

Table 3. Multi-scale urban heat island monitoring spatial resolution adaptation framework.

Scale Level	Spatial Resolution	Primary Data Sources	Typical Sensors	Detection Capabilities	Mitigation Strategy Focus	Research Cases
Urban-Regional Scale	1 km level	MODIS LST products	MODIS Terra/Aqua	- Urban-rural heat differences - Large-scale heat island distribution - Long-term heat island dynamics	- Urban master planning - Regional policy formulation - Sustainable development strategies	- Huang et al. [43] Shanghai heat island climatology; - Zhang et al. [5] global urban comparison: >500 km² cities 4.7 °C vs. 10–50 km² cities 2.5 °C
Neighborhood Scale	30–100 m	Landsat series	Landsat TM/ETM+/OLI-TIRS	- Functional zone thermal environment differences - Land use thermal effects - Building density impacts	- Community thermal vulnerability identification - Green space planning layout - Built environment optimization	Liu and Weng [64] landscape-temperature relationship: 30 m suitable for category analysis, 90 m suitable for landscape analysis
Microclimate Scale	<30 m	High-resolution satellites, Airborne sensors	ECOSTRESS (70 m), TASI (0.6/1.25 m)	- Building-level thermal characteristics - Fine temperature variations - Microenvironment identification	- Building design optimization - Microenvironment improvement - Precision cooling measures	- Liu et al. [58] Shijiazhuang multi-scale analysis: high resolution reveals 7–10 °C temperature differences; - Hulley et al. [23] fine-scale mapping

Table 3. Cont.

Scale Level	Spatial Resolution	Primary Data Sources	Typical Sensors	Detection Capabilities	Mitigation Strategy Focus	Research Cases
Cross-Scale Integration	Multi-resolution integration	Multi-source data fusion, Spatiotemporal fusion techniques	Downscaling techniques, Data fusion algorithms, Machine learning models	- Breakthrough spatiotemporal resolution trade-offs - Multi-level continuous monitoring - Cross-scale consistency	- Comprehensive heat island management - Multi-level planning coordination - Systematic mitigation solutions	- Bechtel et al. [60] 3300 × 6700 m downscaled to 100 m; - Nichol [59] 90 m to 10 m resolution validation

4. Methodological Advances

Based on the technological development achievements elaborated in Section 3, this chapter focuses on analyzing innovative progress at the methodological level, exploring how to effectively transform technological advantages into monitoring capability improvements through algorithm improvements, data fusion, and verification strategy refinement.

4.1. Land Surface Temperature Extraction and Heat Island Intensity Calculation Methods

The core foundation of urban heat island (UHI) research is the precise extraction of land surface temperature (LST), which directly determines the accuracy and reliability of heat island intensity calculations. Through systematic research, the current mainstream land surface temperature extraction algorithms primarily include Mono-Window Algorithm (MWA), Single-Channel Algorithm (SCA), and Split-Window Algorithm (SWA), which perform differently across various urban environments and research scales.

Mono-Window Algorithm (MWA), as one of the most widely applied methods in land surface temperature extraction, is particularly common in Landsat data processing. This algorithm extracts land surface temperature information from a single thermal infrared band by integrating key parameters such as atmospheric transmittance and land surface emissivity. Şekertekin et al. [40] demonstrated that MWA can achieve a root mean square error (RMSE) of less than 2.4 K and a coefficient of determination (R^2) of greater than 0.9, showing impressive accuracy levels. Wang et al. [65] further confirmed the algorithm’s precision potential, achieving an RMSE of 0.43 K under ideal conditions. However, Lei Wang et al. [27] revealed that the algorithm exhibits high sensitivity to input parameters such as atmospheric water vapor content and atmospheric transmittance, a characteristic that may lead to accuracy degradation in humid regions with high water vapor content or structurally complex urban environments.

Split-Window Algorithm (SWA) adopts a more advanced approach, ingeniously utilizing the differences between two or more adjacent thermal infrared bands to compensate for atmospheric effects; it is considered one of the most robust land surface temperature extraction algorithms currently available. Multiple validation studies have demonstrated that SWA can achieve RMSE ranges of 0.51–1.8 K, with Jin et al. [41] attaining a high precision of 0.51 K under mid-latitude summer conditions, and Dave et al. [66] obtaining accuracies of 1.45–1.80 K and 1.10–1.14 K in heterogeneous and homogeneous regions, respectively, proving its excellent adaptability and stability under diverse environmental conditions. Furthermore, the comparative study by Käfer et al. [67] demonstrated that SWA (1.18 K) significantly outperforms single-channel methods (1.6 K), further validating its algorithmic advantages.

In response to the unique complexity of urban environments, researchers have developed a series of improved algorithms specifically adapted to urban three-dimensional structures. Innovative methods such as the Urban Split-Window (USW) algorithm, Extended SW-TES algorithm, Local Linear Forest (LLF), and Improved Mono-Window (IMW) algorithm significantly enhance temperature extraction accuracy in complex urban envi-

ronments by introducing key parameters such as urban geometry and sky view factor (SVF), with RMSE values generally below 1.0 K [42]. Research by Ru et al. [68] further confirms that urban-specific algorithms integrating additional parameters such as urban geometry can more accurately capture the influence of urban canyon effects on temperature distribution, performing particularly well in high-density building areas.

Regarding heat island intensity calculation, there are primarily two main methods: surface urban heat island (SUHI) intensity based on land surface temperature (LST) and Canopy Urban Heat Island (CUHI) intensity based on air temperature. The literature analysis reveals that thermal variation when comparing city centers with the surrounding countryside represents the most commonly used method for calculating SUHI intensity. García [26] innovatively studied the dynamic changes in surface urban heat islands during heat waves using Sentinel-3 data, precisely quantifying the enhancement effect of heat islands under extreme climate conditions through time series analysis. Meanwhile, machine learning-based heat island intensity calculation methods are rapidly emerging. Lyu et al. [28] combined web GIS frameworks with various machine learning models to achieve the high-precision prediction and intensity calculation of urban heat islands at fine spatiotemporal scales. These methods can effectively integrate multi-source remote sensing data, urban morphology information, and human activity characteristics, providing innovative approaches for precise quantification of heat island intensity.

As systematically summarized in Table 4, the selection of land surface temperature extraction and heat island intensity calculation methods should be based on specific research objectives, regional characteristics, and data availability for personalized decisions. For large-scale regional analysis or long-time series research, computationally efficient and robust algorithms such as SWA or MWA may be more suitable; for urban microclimate or fine neighborhood-scale studies, specialized algorithms that consider urban three-dimensional structures can provide higher precision results. Table 4 comprehensively evaluates the principles, applicable conditions, accuracy indicators, advantages, and limitations of various methods, clearly indicating that there is no one-size-fits-all method, and the best choice should be based on specific research contexts and objectives.

Table 4. Comprehensive assessment of land surface temperature extraction and heat island intensity calculation methods.

Method Category	Specific Algorithm	Main Principle	Applicable Conditions	Accuracy Indicators	Advantages	Limitations
Single Window Algorithms	Mono-Window (MWA) Algorithm	Considers parameters such as atmospheric transmittance and land surface emissivity to extract surface temperature from a single thermal infrared band	Landsat series data, moderately complex urban environments	RMSE: 2.39 K [40]	Widely applied, relatively simple calculation	Sensitive to atmospheric water vapor content, decreasing accuracy in humid regions
	Improved Mono-Window Algorithm (IMW)	Introduces urban geometric shape parameters based on traditional MWA	Complex urban environments, high-density building areas	RMSE: <1.0 K [42]	Better consideration of urban canyon effects	Requires additional urban morphology parameters
Split Window Algorithms	Split Window (SWA) Algorithm	Uses differences between two or more adjacent thermal infrared bands to compensate for atmospheric effects	Multi-band thermal infrared data, such as MODIS, VIIRS	RMSE: 0.51–1.8 K [41,66]	Lower sensitivity to atmospheric water vapor estimation errors, good robustness	Requires at least two thermal infrared bands
	Urban Split Window (USW) Algorithm	SWA optimized for urban environments, incorporating urban geometric features	High-density urban areas, especially coastal cities	RMSE: <1.0 K [42]	Stable performance in complex urban environments	Complex algorithm, high parameter requirements
Machine Learning Methods	Local Linear Forest (LLF)	Non-parametric regression method based on machine learning	Data-rich complex urban environments	Data missing	Can handle nonlinear relationships, adapts to complex environments	Requires large training datasets, computationally intensive
	Random Forest Regression	Ensemble learning method, integrating prediction results from multiple decision trees	Multi-source data fusion, highly heterogeneous urban areas	Data missing	Can process multiple data types, strong noise resistance	Black-box characteristics, weaker physical interpretability
Heat Island Intensity Calculation Methods	Urban-Rural Temperature Difference Method	Temperature difference between urban areas and surrounding rural areas	Regions with clear urban-rural boundaries, medium to large cities	Most commonly used method, widely applied in global research	Clear concept, easy to understand and calculate	Inconsistent urban-rural boundary definitions affect result comparability
	Time Series Analysis Method	Analyzes heat island intensity changes at different time scales	Long time series data, seasonal and extreme event studies	Can quantify heat island intensification effects under extreme climate conditions [26]	Can capture dynamic changes, suitable for long-term trend analysis	Requires temporally continuous observation data
	Machine Learning-Based Heat Island Intensity Calculation	Predicts urban heat island intensity by combining multi-source data	Multi-source data available, fine spatiotemporal scale research	Data missing	Can achieve fine-scale urban prediction, considers multi-factor influences [28]	Depends on high-quality multi-source data, complex models

Table 4. Cont.

Method Category	Specific Algorithm	Main Principle	Applicable Conditions	Accuracy Indicators	Advantages	Limitations
Accuracy Assessment Strategies	Ground Truth Validation	Compares satellite LST with ground sensor or meteorological station data	Research supported by ground measurement networks	In areas with canopy coverage > 47%, MAE < 3.74 °C, $R^2 > 0.85$ [20]	Direct comparison, reliable results	Spatial representativeness issues, limited ground measurement networks
	Cross-Platform Validation	Compares data from different satellite sensors	When multi-platform data is available	Uses R^2 , correlation coefficients, and RMSE to evaluate consistency	Extends spatial coverage, provides consistency checks	Sensor differences, atmospheric effects, and calibration issues
	Temporal Validation	Evaluates the consistency and accuracy of UHI measurements over time	Long-term studies, seasonal variation analysis	Usually $R^2 > 0.73$ in winter, R^2 about 0.5 in other seasons [29]	Captures temporal dynamic changes	Need to consider seasonal variations, urban development dynamics, and other factors

Note: RMSE = root mean square error, LST = land surface temperature, UHI = urban heat island, MAE = mean absolute error.

4.2. Integration and Fusion Technologies for Multi-Source Remote Sensing Data

With the rapid development of remote sensing technology, multi-source data integration and fusion have become a key trend in urban heat island research. Different sensor platforms each have unique advantages and limitations; through the clever integration of these data, researchers can achieve “complementary advantages”, significantly enhancing the comprehensiveness and precision of heat island monitoring. As shown in Table 2, emerging data fusion technologies demonstrate how to break through the inherent limitations of single remote sensing systems by combining the characteristics of different platforms.

The core objective of multi-source remote sensing data integration is to address the limitations of single data sources, especially the fundamental trade-off between spatial and temporal resolution. The literature analysis shows that Landsat and MODIS are the most commonly used data sources in heat island research, adopted by 62.5% and 42.5% of studies, respectively, perfectly representing the two poles of spatial resolution advantage and temporal resolution advantage. Platforms such as Sentinel and ASTER provide key supplementary information in specific application scenarios.

4.2.1. Main Types of Fusion Technologies

Spatiotemporal fusion technology has become an effective solution to overcome resolution trade-offs. Liu and Weng [69] ingeniously integrated ASTER (90-m resolution) and MODIS data using the Spatial and Temporal Adaptive Reflectance Fusion Model (STARFM), simultaneously achieving temperature monitoring with high spatial precision and high temporal frequency. The core principle of this method is to use high spatial resolution data to provide fine structural details while utilizing high temporal resolution data to capture dynamic temperature changes. Bird et al. [24] further developed this approach, creating a temperature product with a 100 m spatial resolution and daily updates by fusing Landsat and MODIS data, greatly enhancing the spatiotemporal continuity of heat island monitoring.

Downscaling technology, as another key fusion method, excels in enhancing the spatial details of coarse-resolution data. Bechtel et al. [60] significantly enhanced the spatial details of high-frequency heat island monitoring by using sophisticated downscaling techniques to improve geostationary satellite data from 3300×6700 m to a 100 m resolution while maintaining a high-frequency observation capability of 15 min. These techniques typically rely on land cover information, terrain features, and auxiliary high-resolution data to construct precise relationship models between thermal environments and surface characteristics.

Machine learning methods, with their powerful nonlinear modeling capabilities, demonstrate significant advantages in multi-source data fusion. The analysis found that 20% of relevant studies employed machine learning techniques, including advanced algorithms such as random forests and support vector machines. Comparative research by Zhou et al. [70] confirmed that support vector machine regression outperforms traditional multivariate linear regression and artificial neural networks in heat island data fusion.

These methods can effectively handle complex nonlinear relationships, adapt to highly heterogeneous urban environments, and do not require extensive physical model support. Garzón et al. [25] achieved high-precision prediction and analysis by integrating Landsat and Sentinel-2 data for heat island modeling using machine learning techniques.

Multi-sensor integration, as a direct fusion strategy, comprehensively enhances the comprehensiveness of heat island research by combining the unique observational advantages of different sensors. Mushore et al. [71] innovatively integrated Sentinel-2 and Landsat 8 data, achieving 10 m resolution heat island analysis using panchromatic sharpening technology, particularly suitable for studying the fine-scale influence of roof colors on thermal environments. Shi et al. [61] proposed the Comprehensive Flexible Spatiotemporal Data Fusion (CFSDAF) method, which generates high spatiotemporal resolution urban land surface temperature products by fusing MODIS and Landsat data, providing important technical support for fine-scale monitoring of urban heat islands.

The integration of physical models with remote sensing data provides heat island research with a mechanistic understanding and predictive capabilities. These methods, based on solid physical principles, can deeply explain the underlying mechanisms of heat island formation but also face challenges such as computation intensity and parameter complexity. Rios and Ramamurthy [72] reported an RMSE value of 47.32 W/m^2 for sensible heat flux estimation using a satellite-based model in urban areas, demonstrating the reliable accuracy of such integrated approaches in urban energy balance simulation.

4.2.2. Data Fusion Quality Assessment and Uncertainty Analysis

Based on the development of the various fusion technologies mentioned above, data fusion quality assessment and uncertainty analysis have become key components for ensuring the reliability of fusion results. However, as Zhou et al. [15] pointed out, data and methodological limitations hinder the application of SUHI research, which is precisely a relatively weak area in current research. In multi-source remote sensing data fusion research, quality control mainly involves the following aspects:

Fusion accuracy assessment is the core element of quality control. Different fusion methods show significant differences in accuracy performance. Bird et al. [24] successfully combined the high temporal resolution advantages of MODIS with the high spatial resolution advantages of LANDSAT through temperature-variable DisTrad functions and spatial spectral composition techniques, achieving precision improvements from 2.8°C to 2.1°C in the Paris region application, demonstrating that multi-source data fusion can provide spatiotemporally continuous information while maintaining high accuracy. Yang and Lee's [50] scale separation framework maintained good accuracy across different scenarios through "leave-one-out" cross-validation, proving the robustness of this fusion method.

The impact of surface conditions on fusion quality requires special attention. Surface heterogeneity has significant effects on fusion algorithm performance. Elmes et al. [20] found that in areas with tree canopy coverage exceeding 47%, Landsat 8 land surface temperature maintained consistent relationships with ground measurements ($\text{MAE} < 3.74^\circ\text{C}$, $r^2 > 0.85$), while accuracy decreased under other surface conditions.

Error propagation and uncertainty quantification are important components of fusion quality assessment. Uncertainties in multi-source fusion processes mainly originate from (1) differences in original data quality, including sensor calibration errors and inconsistent atmospheric corrections; (2) spatiotemporal registration errors; (3) algorithm limitations; and (4) surface condition complexity. The cumulative effects of these error sources need to be quantified through systematic uncertainty propagation analysis.

Quality comparisons between fusion methods reveal the advantages and disadvantages of different technical approaches. Spatial-temporal fusion methods such as STARFM

perform well in areas with stable surface coverage, but accuracy may decline in rapidly changing urban environments [73,74]. Machine learning methods often perform better in highly heterogeneous urban areas but require sufficient training data. Scale separation methods show advantages in handling multi-scale urban thermal environment problems, effectively addressing error issues across multiple spatial scales [50].

The establishment of quality control strategies and validation frameworks is crucial for ensuring the reliability of fusion products. Effective quality control should include (1) pre-fusion quality checks to eliminate low-quality original data; (2) fusion process monitoring to assess algorithm performance in real time; (3) post-fusion validation using independent datasets for accuracy assessment; and (4) multi-level validation systems combining ground observations, cross-validation, and physical consistency checks.

Current research still has obvious deficiencies in fusion data quality assessment: the lack of standardized quality assessment indicators, the insufficient analysis of error propagation mechanisms, and the lack of unified standards for quality comparison of different fusion methods. There is an urgent need to establish standardized multi-source fusion data quality assessment systems, including unified accuracy assessment indicator systems, systematic uncertainty quantification methods, comprehensive quality control processes, and quality requirement standards for different application scenarios. Only through such comprehensive quality assurance systems can the scientific reliability and practical application value of multi-source fusion products in urban heat island research be ensured.

4.2.3. Method Selection Principles and Development Trends

Based on the comprehensive analysis of the development status and quality assessment of various fusion technologies mentioned above, Table 5 systematically summarizes the technical characteristics, accuracy performance, and applicable conditions of multi-source remote sensing data fusion methods in urban heat island research, providing a comprehensive comparative basis for scientific method selection under different research scenarios.

Based on the systematic comparative analysis of Table 5, various fusion methods exhibit significant differences and complementary characteristics in technical principles, performance, and application scenarios. The core principles for method selection should be based on the comprehensive consideration of research objectives, spatial scales, data conditions, and accuracy requirements. For large-scale heat island research or long-term trend analysis, spatiotemporal fusion technology, with its high computational efficiency and robust reliability, can effectively break through the spatiotemporal resolution trade-off limitations of single sensors, making it the preferred solution. The simultaneous achievement of the 100-m spatial resolution and daily temporal resolution demonstrated by Bird et al. [24] provides an ideal technical solution for regional-scale heat island dynamic monitoring.

Research on urban microclimates or fine neighborhood scales is more suitable for strategies combining downscaling techniques with multi-sensor integration, as these methods can enhance spatial details while maintaining high observation frequency. The technological breakthrough by Bechtel et al. [60] in enhancing geostationary satellite data from 3300×6700 m to a 100 m resolution laid the foundation for fine-scale heat island monitoring. For detailed thermal pattern analysis in complex urban environments, machine learning methods, with their powerful nonlinear modeling capabilities, excel in handling high heterogeneity and complex spatial relationships. The support vector machine regression advantages confirmed by Zhou et al. [70] and the practical applications by Garzón et al. [25] indicate that such methods often perform better in highly heterogeneous urban areas. For fundamental research focusing on heat island formation mechanisms, physical model integration, with its in-depth explanatory capabilities based on solid physi-

cal principles, can provide more valuable mechanistic insights. Rios and Ramamurthy [72] achieved an RMSE of 47.32 W/m² in their satellite-based sensible heat flux estimation model, confirming the reliability of such physically based approaches for urban energy balance simulation.

Table 5. Comparative analysis of multi-source remote sensing data fusion methods for urban heat island research.

Fusion Method	Technical Approach	Performance Indicators	Applicable Conditions	Spatiotemporal Constraints	Advantages and Limitations
Spatiotemporal Fusion Technology	STARFM (Spatial and Temporal Adaptive Reflectance Fusion Model)	100 m spatial resolution with daily temporal resolution [24]	Stable land cover regions; requires paired high and low-resolution data	Simultaneous high spatial precision and high temporal frequency requirements	+ Enhanced spatiotemporal continuity - Performance may decline in rapidly changing urban environments
Downscaling Technology	Sophisticated downscaling techniques for geostationary satellite data	3300 × 6700 m improved to 100 m resolution, maintaining 15 min observation frequency [60]	Large study areas with diverse land cover types	Relies on land cover information, terrain features, and auxiliary high-resolution data	+ Significantly enhanced spatial details for high-frequency monitoring - Typically rely on auxiliary data to construct precise relationship models
Machine Learning Methods	Random forests, support vector machines, and other advanced algorithms	20% of relevant studies employed machine learning techniques [70] SVM regression outperforms traditional methods	Highly heterogeneous urban environments; data-rich scenarios	Requires extensive training datasets and validation	+ Effectively handle complex nonlinear relationships - Do not require extensive physical model support, but need careful model validation
Multi-sensor Integration	Panchromatic sharpening technology combining different sensor platforms	10 m resolution heat island analysis [71]	Multiple sensor platforms available; studies requiring comprehensive coverage	Combination of unique observational advantages from different sensors	+ Unprecedented continuous monitoring capabilities - Faces challenges such as calibration unification of different sensor data
Physical Model Integration	Integration with urban energy balance models	RMSE: 47.32 W/m ² with R ² = 0.70 for energy flux simulation [72]	Research focusing on heat island formation mechanisms	Based on solid physical principles; faces computation intensity and parameter complexity	+ Deep explanation of underlying mechanisms of heat island formation - Faces challenges such as computation intensity and parameter complexity

Note: All performance indicators and technical specifications are extracted from the studies cited in Section 3.2. RMSE = root mean square error; STARFM = Spatial and Temporal Adaptive Reflectance Fusion Model; SVM = support vector machine.

From the perspective of development trends and technological prospects, multi-source data fusion technology is evolving in three main directions. The development of intelligent fusion algorithms will further enhance adaptability to complex urban environments, with the integration of machine learning and deep learning technologies making the fusion process more automated and precise. Under the trend of multi-dimensional data integration, fusion technology is no longer limited to optical and thermal infrared data but is developing toward integrating multi-dimensional observational data such as microwave, LiDAR, and hyperspectral data. Meanwhile, improvements in real-time processing capabilities will enable fusion technology to better serve dynamic monitoring and early warning applications for urban heat islands.

Quality assurance and standardization are important supports for fusion technology development. Current research still has obvious deficiencies in fusion data quality assessment, and there is an urgent need to establish standardized multi-source fusion data quality assessment systems, including unified accuracy assessment indicators, systematic uncertainty quantification methods, comprehensive quality control processes, and quality requirement standards for different application scenarios. In practical applications, re-

searchers should avoid the “one-size-fits-all” method selection and should comprehensively consider specific research objectives, data availability, computational resource conditions, and accuracy requirements, adopting the most suitable fusion strategies or combinations of multiple methods. With continuous innovation in new-generation remote sensing technologies and data processing algorithms, multi-source data fusion technology will play an increasingly important role in urban heat island research, providing more comprehensive and precise scientific evidence for urban sustainable development and climate adaptation planning. However, achieving this goal still requires addressing technical challenges such as the calibration unification of different sensor data, algorithm complexity, computational requirements, and error propagation. Future research should focus on optimizing fusion algorithms’ adaptability to complex urban environments while developing more effective validation methods to ensure the reliability and accuracy of fusion results.

4.3. Accuracy Assessment and Validation Strategies

Accuracy assessment and validation are crucial steps in ensuring the scientific reliability of urban heat island remote sensing research. As remote sensing technologies and data processing methods become increasingly diverse, constructing a systematic and comprehensive validation framework has become particularly important. As shown in Table 4, validation methods for urban heat island research primarily fall into three major categories: ground truth validation, cross-platform validation, and temporal validation, complemented by statistical indicators, spatial analysis, and temperature comparison methods for comprehensive accuracy assessment. These strategies each have their own advantages and limitations; best practice involves integrating multiple validation methods to evaluate the reliability of research results from all dimensions.

Ground truth validation, as the most fundamental validation means, assesses the accuracy of remote sensing results by directly comparing satellite-derived surface temperatures with ground sensor or meteorological station data. Research by Elmes et al. [20] found that in areas with tree canopy cover exceeding 47%, temperatures obtained by Landsat 8 showed high consistency with ground measurements (mean absolute error < 3.74 °C, $R^2 > 0.85$). However, this method faces challenges such as insufficient spatial representativeness, temporal mismatches, and limited coverage. Ru et al. [42] found that traditional algorithms often overestimate surface temperatures in high-density building areas, emphasizing the special difficulty of precise validation in complex urban environments.

By comparing observational data from different satellite sensors, cross-platform validation is particularly valuable when ground measurement resources are limited or when multi-scale consistency needs to be evaluated. Researchers often use MODIS and Landsat data for cross-validation, using indicators such as R^2 , correlation coefficients, and RMSE to assess result consistency [75]. The advantages of this method lie in expanding spatial coverage, providing systematic checks, and enhancing temporal flexibility, but it simultaneously faces challenges such as sensor differences, atmospheric effects, and inconsistent calibration standards [77?].

Temporal validation focuses on evaluating the temporal consistency and stability of heat island measurements, typically comparing short-term satellite observations with long-term climate data or analyzing seasonal and diurnal patterns of heat island intensity. Research by Wan et al. [29] showed that temperature retrieval algorithms typically perform excellently in winter ($R^2 > 0.73$) but with lower accuracy in other seasons (R^2 around 0.5), revealing the profound influence of seasonal factors on validation results. Temporal validation needs to comprehensively consider seasonal variations, diurnal fluctuations, long-term trends, and climate variability while addressing complex challenges such as data continuity gaps, urban dynamic development, and climate change impacts.

The statistical indicator system for accuracy assessment is the core pillar of validation work. Root mean square error (RMSE) quantifies overall prediction bias, mean absolute error (MAE) evaluates the average magnitude of errors, coefficient of determination (R^2) measures consistency between datasets, Mean Bias Error (MBE) identifies systematic tendencies of overestimation or underestimation, Spearman correlation assesses nonlinear relationships, and Mean Absolute Percentage Error (MAPE) is applicable for comparing errors across different scales. As research deepens, scholars increasingly emphasize using multiple complementary indicators to build comprehensive assessment systems and interpret results within specific spatial and temporal contexts.

Spatial analysis techniques targeting the highly spatial heterogeneity of urban heat islands assess how effectively satellite imagery represents geographic differences in thermal environments. Spatial correlation analysis, buffer analysis, land cover-based stratified validation, and spatial downscaling assessment each have their own characteristics suitable for different validation contexts. Nichol [59] achieved a high correlation ($R^2 = 0.71\text{--}0.72$) by comparing emissivity-modulated image temperatures with 18 in situ measurement points, demonstrating the excellent effect of spatial validation in identifying microscale temperature patterns. Best practices include using high-resolution land cover data for stratified validation, applying geostatistical techniques to analyze spatial patterns, considering the influence of urban three-dimensional structures, and integrating multi-source spatial datasets to enhance validation accuracy.

Temperature comparison methods focus on the complex relationship between surface temperature and air temperature, as well as the performance differences of urban heat islands under various spatiotemporal conditions. LST–air temperature regression analysis, diurnal temperature curve comparison, seasonal temperature pattern comparison, and urban–rural temperature gradient analysis each have their features, with specific key parameters and reliability indicators. These methods need to comprehensively consider factors such as surface–air temperature dynamic relationships, temporal variation patterns, urban–rural gradient characteristics, and microclimate effects. Best practices include deploying dense ground temperature sensor networks to capture spatial variations in air temperature, conducting systematic comparisons across multiple time scales, performing stratified comparisons by land cover type, and considering the influence of measurement height and atmospheric conditions on comparison results.

Overall, effective accuracy assessment and validation strategies should be based on specific research objectives and data characteristics, adopting a diversified, systematic methodological framework. Building on conventional statistical indicators, combined with spatial analysis techniques and temperature comparison methods, research should comprehensively evaluate the reliability and applicability of urban heat island remote sensing. Complementary relationships exist between different validation methods: ground truth validation provides direct reference standards, cross-platform validation extends spatial coverage, and temporal validation captures dynamic change processes. Future research should strive to develop standardized validation protocols to promote comparability between different studies while strengthening long-term validation research to systematically evaluate the temporal stability and predictive reliability of heat island models. Particularly for microscale heat island effects in complex urban environments, more refined validation techniques need to be developed, combining high-resolution ground observation networks and three-dimensional urban models to enhance the precision and representativeness of validation.

5. Comparative Study of Heat Island Characteristics

This chapter reveals the scientific discoveries of heat island research supported by technological development and methodological innovation through a systematic comparison of heat island characteristics between metropolitan areas and small-medium cities, as well as an in-depth analysis of the relationship between urban morphology and thermal environment, providing a theoretical foundation for practical applications.

5.1. Comparison of Heat Islands Between Metropolises and Small-to-Medium City Sections

Metropolises and small-to-medium cities exhibit significant differences in urban heat island (UHI) effects, reflecting the complex relationships between urban scale, morphology, and the thermal environment. Metropolitan areas typically demonstrate more intense heat island effects due to their high degree of urbanization, dense building clusters, and extensive impervious surfaces. A systematic literature analysis reveals particularly pronounced heat island phenomena in rapidly urbanizing regions of Asia: Isfahan, Iran, experienced a surface temperature increase from 36.08 °C to 43.68 °C [78]; Chennai, India, saw a 6.53 °C surface temperature rise over the past decade [79], and Beijing projects a 7.5 °C surface temperature increase by 2050 [80]. These temperature changes typically accompany urban expansion processes, as documented by Yang et al. [81], who recorded Changchun's impervious surface expansion from 143.15 km² in 1984 to 577.45 km² in 2014, resulting in dual growth in heat island intensity and extent. Li et al. [82] observed that after Shanghai's built-up area increased by 219.50%, the heat island effect expanded radially from the center outward, forming a broader thermal impact zone.

While small-to-medium cities generally exhibit lower heat island intensities than metropolises, they display more complex and diverse patterns. Zhang et al.'s [5] global study indicated that cities of 10–50 km² averaged a summer heat island intensity of approximately 2.5 °C, compared to 4.7 °C for cities exceeding 500 km². However, heat island intensity can reach extreme levels in small-to-medium cities—Amorim et al. [3] demonstrated that medium-sized cities in both tropical and temperate climates can experience extreme urban heat island intensities exceeding 6 °C, which are often overlooked compared to the focus on major metropolitan areas; Burnett and Chen [83] recorded intense heat island effects of 6.55–8.09 K in a Canadian small city during summer, demonstrating that city size is not the sole determinant of heat island intensity.

Regarding spatial distribution patterns, metropolitan heat islands typically exhibit a “concentric circle” model, decreasing from the city center toward the periphery [82,84], reflecting the spatial gradient of urban density and activity intensity. In contrast, small-to-medium cities often display more irregular heat island patterns, significantly influenced by local geographical factors [85]. Interestingly, Ambinakudige [85] discovered that some small-to-medium cities exhibit a “reverse pattern”, where urban core temperatures are lower than peripheral areas, a phenomenon potentially linked to the distribution of water bodies and green spaces in core areas.

Methodologically, metropolitan studies predominantly employ established single-window or split-window algorithms for surface temperature extraction, calculating heat island intensity using urban–suburban temperature difference methods. Liu et al.'s [58] innovative research in Shijiazhuang demonstrated the advantages of multi-platform data integration, cleverly combining Landsat TM with airborne thermal infrared sensors to achieve multi-scale heat island analysis. In contrast, heat island monitoring in small-to-medium cities faces distinct challenges, particularly regarding spatial resolution adaptation. Due to the limited urban scale, internal thermal environment differences typically exist at finer spatial scales. Liu and Weng [64] noted that a 30-m resolution is suitable for category-based thermal environment analysis, which is especially important for small-to-medium city re-

search. Additionally, small-to-medium cities typically have higher mixed pixel proportions in boundary areas, increasing temperature extraction uncertainty. Furthermore, ground observation networks in small-to-medium cities are often less comprehensive than those in major cities, with sparse meteorological stations, making validation more challenging.

Notably, small-to-medium cities exhibit significant “spillover effects.” Burnett and Chen [83] found that the heat island footprints of small suburban centers extend 1.4 times the city’s own area, indicating that even smaller cities can generate thermal influences on surrounding rural environments far beyond their physical boundaries.

The relationship between city size and heat island intensity demonstrates nonlinear characteristics. Li et al. [30] found (through panel analysis) that heat island intensity increases by approximately 0.7 °C for each doubling of urban area. However, research has also documented exceptional cases, such as Fan et al.’s [86] observation of a pronounced “oasis effect” in desert small cities, where urban areas were 7.8 °C cooler than the surrounding environments. These findings indicate that local climatic conditions and urban morphological characteristics play important regulatory roles in the relationship between city size and heat islands.

5.2. Key Findings on the Relationship Between Urban Form and Heat Islands

The relationship between urban form and heat island effects is complex and intricate, with remote sensing research providing rich evidence revealing how urban structure, layout, and components shape the urban thermal environment.

Building density and three-dimensional urban structure are core drivers of heat island formation. Multiple studies have demonstrated a significant positive correlation between building density and land surface temperature. Lu et al. [31] analyzed Xi’an’s thermal environment using the Percentage of Landscape (PLAND) index, finding that building density’s impact on temperature far exceeded green space’s moderating effect. This finding is echoed in Otaghsara and Arefi’s [87] research in Santa Rosa, CA, USA, where they recorded a strong positive correlation between building volume and surface temperature (coefficient reaching 6.654), quantifying the significant impact of three-dimensional urban structure on the thermal environment.

The urban canyon effect, as a typical thermal characteristic of high-density building areas, alters heat exchange mechanisms between the surface and atmosphere while significantly affecting solar radiation reception and long-wave radiation release patterns. Wang et al. [2] observed in Phoenix that the night-time heat island affected area (410 km²) exceeded the daytime coverage, with the maximum night-time temperature increase reaching 5.35 °C. This night-time heat island enhancement phenomenon is closely related to heat-trapping within urban canyons.

Green spaces and water bodies, acting as urban “cool islands”, play crucial roles in regulating the thermal environment. Multiple studies conducted in different urban environments have confirmed that vegetation coverage generally exhibits negative correlations with land surface temperature, but the correlation strength and stability vary according to geographical regions and conditions. Puche et al. [88] recorded a significant 6.7 °C temperature difference between natural areas and built-up zones; Kumar et al. [84] observed that green-covered areas were 1.2 °C cooler than impervious surfaces; and Osei et al. [89] quantified the strong negative correlation between vegetation cover and surface temperature (R value of −0.77) in Sunyani, Ghana. Vegetation’s cooling effect exhibits distinct seasonal variation patterns—Zhao et al. [90] found that vegetation in Beijing provides cooling effects in summer while potentially producing warming effects in winter; Zoran et al. [91] noted that Bucharest’s vegetation cooling effect significantly enhanced during summer heatwaves, indicating vegetation’s special regulatory value during extreme heat events.

Water bodies typically exhibit relatively lower land surface temperatures and smaller seasonal fluctuations in urban environments [92], but their specific regulatory effects are significantly influenced by factors such as water body size, depth, location, and surrounding environment. Alhawiti and Mitsova [93] found that areas near water bodies had noticeably lower temperatures than other urban areas, with water body cooling effects radiating to surrounding areas and forming extended cooling impact zones. Ambinakudige's [85] findings are particularly noteworthy, noting that urban core areas with water bodies and abundant vegetation had lower temperatures than peripheral regions, challenging the traditional concentric heat island model and revealing the possibility of inner-city cool island effects.

The temporal and spatial variation characteristics of heat islands reflect complex interactions between natural and anthropogenic factors. Seasonal variation is a prominent feature of heat island effects, with multiple studies consistently showing summer as the peak period for heat island intensity [88,94,95]. Zeng et al. [95] recorded heat island intensity differences between seasons reaching 4–6 °C, quantifying the significant magnitude of this temporal dynamic variation. Diurnal variations in heat island intensity reveal fundamental differences between urban and natural surface thermal characteristics. Wang et al. [2] observed that the night-time heat island-affected area in Phoenix (410 km²) far exceeded the daytime warming zones, with all areas showing warming trends and maximum temperature increases reaching 5.35 °C, revealing the unique characteristics of night-time heat release patterns.

Regarding spatial distribution patterns, Wang et al. [2] captured the spreading trend of Phoenix's heat island hotspots from traditional city centers toward newly developed suburbs, vividly demonstrating urban sprawl's spatial restructuring of the thermal environment; Li et al. [82] observed Shanghai's heat island radiating outward from the core area, forming a broader thermal influence range. Long-term trend studies confirm that heat island effects typically show intensifying patterns with urban development. Yang et al. [81] tracked Changchun's impervious surface expansion process, finding that heat island effects not only increased in intensity but also strengthened in correlation with impervious surfaces over time; Ernest et al. [96] recorded impervious surfaces surging from 9 km² in 1990 to 82 km² in 2015, accompanied by a synchronous growth in heat island intensity.

Table 6 comprehensively summarizes the quantitative analysis results of major urban morphological elements' relationships with the thermal environment, clearly showing the specific influence intensity and mechanisms of key morphological elements such as impervious surfaces, building density, vegetation cover, water bodies, and urban scale on heat island effects. These precise quantitative relationships provide a solid scientific foundation for urban planning and heat island mitigation strategies. While the impacts of different urban morphological elements on the thermal environment show certain regional variations, the quantitative relationships shown in Table 6 reveal some universal cross-regional patterns. These findings not only help understand the basic physical mechanisms of urban heat island formation but also lay the foundation for analyzing regional differences and universal patterns in the next section. Through the systematic comparison of these quantitative indicators, key elements universally affecting heat island effects globally can be clearly identified, as well as factors showing significant regional specificity, providing a basis for precise heat island management.

Table 6. Quantitative analysis of the relationship between urban morphology and the thermal environment.

Urban Morphology Elements	Impact Relationship	Quantitative Indicators	Case Studies	Regional Differences	Planning Implications
Impervious Surface	Significant positive correlation with LST	NDISI correlation: 0.59–0.97; Explains > 60% LST variance	Lu et al. [31], Zhang et al. [5]	Major driving factor across climate zones globally	Controlling impervious surface is core UHI mitigation strategy
	Enhanced with urban expansion	Impervious surface: 143.15 km ² to 577.45 km ² , with synchronous UHI increase	Yang et al. [81], Changchun study	Common global phenomenon	Plan mitigation with urban expansion
	Long-term cumulative effect	Impervious surface: 9 km ² to 82 km ² (1990–2015), with UHI increase	Ernest et al. [96]	More evident in developing regions	Long-term planning should include thermal assessment
Building Density and 3D Structure	Strong positive correlation with LST	Building Volume coefficient: 6.654	Otaghsara Arefi [87], and Santa Rosa	Positive correlation across climate zones	Zoning control improves thermal environment
	Urban canyon effect	Night-time UHI area (410 km ²) exceeds daytime warming zones (176 km ²), max increase 5.35 °C	Wang et al. [2], Phoenix	Varies by urban morphology	Design ventilation corridors to prevent heat trapping
	Heat island spatial pattern	Concentric patterns decreasing from center to periphery	Kumar et al. [84], Li et al. [82]	Smaller cities show irregular patterns, more local influence	Include urban scale in thermal planning
Vegetation Cover	Negative correlation with LST	Coefficient: −0.77; Natural areas 6.7 °C cooler; Green areas 1.2 °C cooler than impervious	Osei et al. [89], Puche et al. [88], Kumar et al. [84]	Cooling effect across regions and scales	Urban green space is effective UHI mitigation
	Seasonal variations	Summer cooling, possible winter warming	Zhao et al. [90], Beijing	More seasonal differences in temperate cities	Consider seasonal characteristics in planning
	Enhanced during heat waves	Significantly enhanced cooling during summer heat waves	Zoran et al. [91], Bucharest	Global phenomenon with climate variations	Vegetation crucial for extreme heat response
Water Bodies	Significant cooling effect	Lowest LST, less seasonal variation	Su et al. [92], Alhawiti Mitsova [93] and	Cooling extends to surrounding areas	Water system planning improves thermal environment
	Pattern modification	Cities with water bodies have lower core temperatures	Ambinakudige [85]	Challenges traditional concentric pattern	Water layout can alter thermal patterns
Urban Size and UHI Intensity	Positive correlation	Cities > 500 km ² : 4.7 °C UHI; 10–50 km ² cities: 2.5 °C	Zhang et al. [5]	Common global phenomenon	Metropolitan areas need systematic mitigation
	Area expansion effect	Doubling area increases temperature by 0.7 °C	Li et al. [30]	Found across different regions	Consider growth and thermal changes together
	Exceptional cases	“Oasis effect”: desert cities 7.8 °C cooler than surroundings	Fan et al. [86]	Unique to arid regions	Adapt strategies to regional climate
Spatiotemporal Variation	Seasonal variations	Seasonal differences: 4–6 °C; US cities: summer 4.3 °C, winter 1.3 °C	Zeng et al. [95]	More variation in temperate cities; Cold regions: winter UHI > summer	Consider seasonal differences in mitigation
	Diurnal variations	Night-time UHI area (410 km ²) larger than daytime zones (176 km ²), max increase 5.35 °C	Wang et al. [2]	Different patterns across regions	Night-time UHI needs special attention
	Long-term evolution	UHI strengthens with development; Global UHI growth: 0.156 °C/decade	Yang et al. [81], Li et al. [4]	More intense in rapidly urbanizing areas	Long-term planning should predict thermal changes

Note: LST = land surface temperature; NDISI = Normalized Difference Impervious Surface Index; UHI = urban heat island.

5.3. Regional Differences and Universal Patterns

Based on the systematic literature review methodology of this study (detailed in Section 2), global urban heat island studies across different regions reveal complex phenomena that both exhibit regional characteristics and follow common patterns, providing valuable perspectives for comprehensively understanding heat island mechanisms through such comparative analysis.

Climate zones significantly shape heat island characteristics. Amorim et al.'s [3] research found that heat island effects in tropical environments are more pronounced and prevalent than in temperate cities, with extreme heat island intensities in medium-sized cities exceeding 6 °C. This phenomenon may stem from tropical regions' year-round high temperature and high humidity climate background, where urbanization-induced reduction in evapotranspiration and anthropogenic heat emissions produce more intense temperature effects. Arid regions exhibit a unique "oasis effect." Fan et al. [86] observed up to 7.8 °C reverse temperature differences in desert cities, where the urban areas are cooler than surrounding environments, forming an "urban cool island" phenomenon, primarily attributed to the contrast between artificial irrigation and vegetation maintenance within cities vs. the surrounding arid environment.

Temperate cities' heat islands show distinct seasonal fluctuations. Yang et al. [97] recorded summer surface heat island intensities of 6.83 °C compared to 1.55 °C in winter in the temperate snow climate city of Changchun, demonstrating significant seasonal differences. Biome background also influences heat island intensity, with Zhang et al. [5] observing average urban heat islands of 3.8 °C in forest biomes, significantly higher than the 1.9 °C in grassland-shrubland biomes. Cold climate zone cities demonstrate anomalous seasonal patterns. Miles and Esau [98] found that northern West Siberian cities' heat island intensities were higher in winter (1.4–1.9 °C) than in summer (0.3–1.1 °C), with this winter enhancement primarily related to large amounts of anthropogenic heat released during heating periods.

Beyond heat island research in Asia and North America, urban heat island studies in European regions also exhibit distinctive regional characteristics. Costanzini et al. [99] provide a typical case study through their long-term temperature trend analysis of Italy's Po Valley, successfully quantifying the specific contribution of the urban heat island effect to temperature changes by comparing historical data from an urban site (Modena Observatory) and a rural site (Mount Cimone Observatory). The research found that during the period of 1981–2018, the warming trend of maximum temperatures reached 0.84 °C·decade^{−1} in Modena and 0.62 °C·decade^{−1} at Mount Cimone; the minimum temperature trends were 0.77 °C·decade^{−1} and 0.80 °C·decade^{−1} respectively, almost double the increase during the 1951–2018 period. It is particularly noteworthy that, through a comparative analysis of the two sites, this study precisely quantified the contribution of urbanization to extreme temperature events in Modena: the urban heat island effect accounted for up to 65% of the increase on warm nights (TN90p), 57% of the increase on tropical nights, 37% of the decrease on frost days, and 39% of the decrease on cool nights. These findings not only confirm the significant presence of the urban heat island effect in medium-sized European cities but also reveal the amplifying effect of urbanization on local temperature changes in the context of global climate change, providing a methodological reference for separating the relative contributions of global climate change and urban heat island effects. Compared to research in Asia and North America, European urban heat island studies place greater emphasis on long time series analysis and changes in extreme climate indices, reflecting differences in regional research methodologies.

Despite obvious regional differences, research has also revealed a series of cross-regional universal patterns. Impervious surface proportion, as the dominant driving factor of heat island intensity, has been validated in multiple studies analyzed in this review. Zhang et al. [5] found that impervious surfaces explain over 60% of surface temperature variance in forest-region cities; Lu et al. [31] observed correlation coefficients as high as 0.59–0.97 between surface temperature and the impervious surface index, quantifying the stability of this relationship.

The cooling effects of vegetation and water bodies have been confirmed in most studies, but their effectiveness varies significantly due to geographical conditions and management

levels. However, specific cooling effects are modulated by multiple factors: Puche et al. [88] recorded a temperature difference of 6.7 °C between natural areas and built-up areas, while Kumar et al. [84] observed only a 1.2 °C cooling effect from green spaces. Vegetation cooling effects also show pronounced seasonality and climate dependency, with Zhao et al. [90] finding (in Beijing) that vegetation provides cooling in summer but may cause warming in winter; Zoran et al. [91] observed significantly enhanced cooling effects from the vegetation in Bucharest during summer heat waves. Water body temperature regulation is similarly influenced by scale and environmental factors, with Jiang et al. [100] recording a 2.90 °C cooling effect near rivers in Shanghai, but the effective range and intensity vary according to water body size, depth, and surrounding land use. Furthermore, factors such as human activity intensity, irrigation management levels, vegetation maintenance conditions, and urban density further complicate the actual cooling performance of vegetation and water bodies, requiring assessment based on local conditions in specific applications.

The trend of heat island intensity increasing with urban scale has shown consistency across multiple studies. Zhang et al.'s [5] global study demonstrated that large cities exceeding 500 km² average a heat island intensity of 4.7 °C, while small cities of 10–50 km² average only 2.5 °C. Li et al. [30] quantified this relationship, finding that surface temperature increases by approximately 0.7 °C for each doubling of urban area. Although specific values fluctuate across different regions, the positive correlation between urban scale and heat island intensity has demonstrated strong consistency in existing studies.

Seasonal variation has emerged as a common characteristic of heat islands in the analyzed studies, with most research indicating the highest intensity during summer. This seasonal pattern spans different climate zones and city types despite variations in specific intensity and magnitude of change. The positive correlation between building density and heat island intensity has been validated in multiple studies, showing strong consistency.

In summary, urban heat islands, as complex urban climate phenomena, are influenced by the combined effects of urban morphology, land cover, human activities, and the regional climate background. Understanding these factors and their interactions is crucial for developing effective heat island monitoring and mitigation strategies. Continuous advances in remote sensing technology, particularly the application of high-resolution data, multi-source fusion, and advanced algorithms, provide us with more powerful tools, enabling a more comprehensive and in-depth understanding of the multi-dimensional complexity of urban thermal environments, with this offering scientific support for urban climate management.

6. Heat Island Mitigation Applications

This chapter transforms the achievements of technological development and methodological innovation described previously into practical application value. The focus is on analyzing the specific contributions of remote sensing technology in heat island mitigation practices, embodying the complete realization of the “Technology-Method-Application” framework.

6.1. Thermal Vulnerability Zone Identification and Green Space Planning

Thermal vulnerability zones are special urban areas with temperatures significantly higher than surrounding environments while simultaneously exhibiting socioeconomic vulnerability. These zones should be prioritized for heat island mitigation strategies. By cleverly integrating satellite-derived surface temperature data, urban morphology indicators, vegetation cover information, and socio-demographic data, remote sensing technology can precisely identify these areas, providing a scientific basis for targeted planning interventions.

Thermal vulnerability zone identification centers on remote sensing data integration and multi-dimensional analysis. Elmarakby and Elkadi [32] innovatively combined remote

sensing, GIS, and statistical analysis to develop a Heat Risk Index (HRI), revealing that approximately 30% of Manchester's area is severely affected by heat island effects, with the city center being particularly vulnerable. As a mainstream method for identifying thermal vulnerability zones, the Heat Vulnerability Index (HVI) was applied by Hulley et al. [23] to integrate ECOSTRESS and MODIS data with socio-demographic indicators, successfully pinpointing the five most vulnerable communities at the urban census block level. Palanisamy et al. [101] analyzed 21 years (2000–2021) of MODIS temperature data for India's National Capital Region, finding severe internal urban thermal environment inequality, with over 75% of vulnerable populations concentrated in high-temperature zones comprising only 10% of the city area, revealing high spatial clustering of thermal environments and social vulnerability.

Remote sensing research clearly reveals the key characteristics and formation mechanisms of thermal vulnerability zones. Zhu et al. [102] found that affluent communities in Phoenix generally enjoy lower temperature environments, whereas low-income communities endure higher heat exposure, quantifying the close correlation between socioeconomic status and thermal environment quality. Todeschi et al. [103] developed a Heat-related Elderly Risk Index (HERI) in Padua, Italy, innovatively integrating surface temperature, population density, and elderly population proportion data and identifying approximately 20% of urban areas facing high heat risk; this provides decision support for the heat protection of specific populations. Wong et al. [104] conducted in-depth research in Hong Kong using logistic regression and spatial autocorrelation analysis, finding that, among 287 planning units, more than half of the areas with vulnerable populations face significant heat island impacts, highlighting the widespread distribution of heat risk.

Remote sensing technology plays a quantitative guiding role in green space planning, supporting the scientific layout of different types of green interventions. Jha et al. [21] used Landsat 8 data to analyze Delhi urban parks' cooling effects, recording significant temperature differences of up to 8.28 °C, confirming the powerful cooling capacity of centralized green spaces. Asadi et al. [33] confirmed that green roofs in Austin could achieve average temperature reductions of 1.96 °C, providing quantitative evidence for building-level heat island mitigation. In long-term research for Beijing, Yao et al. [105] revealed that a 10% increase in forest cover could reduce summer surface temperature by 0.53 °C and winter surface temperature by 1.11 °C, quantifying the seasonal benefits of vegetation increase. Jiang et al. [100] used high-precision remote sensing data to capture cooling effects of up to 2.90 °C near Shanghai rivers, confirming the synergistic cooling value of blue-green infrastructure.

The combination and strategic placement of different green space types can significantly optimize overall cooling effects. Xu et al. [106] provided quantitative evidence from a study of major Chinese cities, finding that the combined strategy of increasing green space coverage and optimizing spatial distribution can generate synergistic cooling benefits of 0.034–0.341 °C, confirming the optimization value of integrated deployment. Yan et al. [107] used high-resolution SPOT 6 and Landsat 8 data to analyze Beijing's green space effects, recording temperature reductions up to 5.1 °C and verifying the significant value of large urban green spaces. Liu et al.'s [108] innovative research revealed complex spatial heterogeneity relationships between Beijing's urban green spaces and surface temperature, providing a scientific basis for refined green space layout. When researching Beijing's sub-center, Cao et al. [109] quantitatively showed that a 10% increase in vegetation cover brings 0.58–0.68 °C cooling, providing specific guidance parameters for urban planning. Li et al. [110] further confirmed that multi-layered urban green spaces (combined configurations of trees, shrubs, and grasslands) provide stronger cooling effects than single green space types, while Shi et al. [111] demonstrated that the combined cooling effect of urban green and blue spaces is greater than the sum of their individual effects accord-

ing to a Chongqing case study, collectively validating the scientific value of synergistic deployment strategies.

The integration of remote sensing and socioeconomic data provides a comprehensive decision-support framework for urban green space planning. During extensive research on 497 US urban areas, Chakraborty et al. [7] found that communities with higher incomes and white populations generally enjoy lower heat island intensity, quantifying the widespread existence of environmental inequality. Sabrin et al. [112] innovatively developed an Environmental Risk Impact Index (ERII) and Social Vulnerability Index (SVI) in Camden, New Jersey, precisely identifying specific communities facing dual risks of heat islands and air pollution, providing tools for environmental justice-oriented planning. Gupta et al. [113] found through heat risk zoning that approximately 690,000 people face moderate humid heat stress, and 210,000 suffer from intense heat stress impacts, providing a quantitative basis for health risk-based green space priority layouts.

6.2. Urban Materials and Surface Property Optimization

Urban surface albedo is one of the key factors affecting the urban thermal environment, with remote sensing technology providing strong support for its assessment and improvement. Smith et al. [114] found in long-term remote sensing observations of seven US cities that each 1% increase in tree cover reduces temperature by 0.089 K, scientifically validating the actual effectiveness of increasing albedo in heat island mitigation.

High-albedo material application has become an important urban heat island mitigation strategy. In researching Abu Dhabi, UAE, Đorđević [34] showed that high-albedo roofs can significantly reduce surface temperature by up to 4.5 °C, highlighting the special value of high-albedo materials in tropical desert climate zones. By using urban surface analysis using high-resolution WorldView3 remote sensing data, Despini et al. [115] revealed that selectively increasing the albedo of clay tile pitched roofs, dark roofs, and parking areas can significantly reduce urban heat island intensity while providing substantial energy savings in air conditioning systems and cost reductions, with a relatively quick return on investment. Seeberg et al. [62] combined Landsat satellite data with visual information in Stuttgart research, further confirming that, from 2004–2008 to 2016–2020, through green roofs and albedo optimization, the city successfully reduced overall surface heat island intensity by 1.4 °C.

The influence of urban three-dimensional structures on the thermal environment is equally important. Remote sensing technology combined with Digital Elevation Models (DEMs) and Light Detection and Ranging (LiDAR) data provides new dimensions for analyzing urban morphology and thermal environment relationships. This is particularly useful for urban canyons, which are street spaces formed by buildings, the geometry of which directly affects thermal environmental characteristics. Research shows that well-designed ventilation corridor layouts can effectively improve urban ventilation conditions, accelerate heat dissipation, and reduce heat island effects.

Water bodies and permeable surfaces play unique roles in urban thermal environment regulation. Du et al. [116] used remote sensing data for quantitative research on Shanghai, systematically analyzing urban green spaces and water bodies' regulatory effects on the thermal environment. Fall et al. [35] used Landsat 8 data for factor and geospatial analysis in Birmingham, Montgomery, and Auburn-Opelika, Alabama, clearly revealing strong positive correlations between surface temperature and impervious surfaces; they also showed significant negative correlations with vegetation index (NDVI), providing strong evidence for water bodies and permeable surfaces' thermal regulatory functions.

To systematically evaluate different heat island mitigation strategies' comprehensive performance, Table 7 comprehensively summarizes the remote sensing evaluation results

of various mitigation measures based on vegetation, materials, water bodies, and their integrated applications. This comparative framework shows each strategy's characteristics, from cooling effects and implementation conditions to socioeconomic benefits in multiple dimensions, providing a scientific decision-making basis for urban planners and policymakers. As shown in Table 7, various heat island mitigation strategies each have advantages: vegetation-based strategies typically achieve more significant cooling effects and provide multiple ecological benefits but require larger space investment and long-term maintenance; material-based strategies are relatively simple to implement with lower costs, although unit cooling effects may be limited; water-based strategies excel in forming urban “cool islands” while featuring landscape beautification and recreational functions; comprehensive strategies show significant advantages in balancing costs and benefits through optimizing combinations of various measures. Translating these research findings into practical applications requires reliance on remote sensing-based scientific decision support systems to ensure the precision and effectiveness of heat island mitigation measures.

Table 7. Comprehensive analysis of remote sensing assessment results for urban heat island mitigation strategies.

Mitigation Strategy	Specific Measures	Cooling Effect	Implementation Conditions and Limitations	Socioeconomic Benefits	References
Vegetation-Based Mitigation Strategies					
Urban Parks and Green Spaces	Urban parks and green spaces	Urban parks can achieve cooling effects of up to 8.28 °C	Requires sufficient land area; requires good water source conditions; larger green spaces have the best effect	Enhances urban aesthetics and livability; increases recreational space; improves air quality; reduces air conditioning energy consumption	Jha et al. [21]
Urban Green Spaces	Urban Forests	10% increase in forest cover can reduce summer temperature by 0.53 °C and winter by 1.11 °C	Requires large land areas; plant species selection must adapt to local climate; requires long-term maintenance	Carbon sequestration and air purification; increases biodiversity; reduces urban noise pollution	Yao et al. [105]
	Green Corridors and Riparian Vegetation	Riverside green corridors can achieve cooling effects of up to 2.90 °C; forms urban “cool island” areas; facilitates cold air circulation	Requires planning along water bodies or road systems; needs integration with overall urban planning	Improves urban ventilation conditions; provides ecological corridors; enhances urban flood control capacity	Jiang et al. [100]
Green Roofs	Multi-layer structure systems, including waterproofing layer, drainage layer, filter layer, growing medium, and vegetation	In the study area, can reduce surface temperature by an average of 1.96 °C in summer	Requires assessment of roof load-bearing capacity; needs waterproofing layer and drainage system; requires regular maintenance and irrigation	Extends roof lifespan; increases biodiversity; reduces building energy consumption; rainwater retention and purification	Asadi et al. [33]
Material-Based Mitigation Strategies					
High-Albedo Roofs	Using high-reflectivity coatings or materials (white or light-colored) to increase the ability to reflect sunlight	Research in Abu Dhabi, UAE shows surface temperature reduction of up to 4.5 °C	Suitable for flat roofs or low-slope roofs; requires regular cleaning and maintenance to maintain high reflectivity	Relatively low implementation cost; simple maintenance; reduces air conditioning energy consumption; extends roof service life	Dorđević [34]
Permeable Pavements	Using permeable concrete, permeable bricks, or gravel surfaces to increase rainwater infiltration	Research reveals strong positive correlation between impervious surfaces and surface temperature, indicating effective temperature reduction	Suitable for sidewalks, parking lots, and other low-traffic areas; requires regular cleaning to prevent clogging	Reduces urban runoff; replenishes groundwater; reduces urban heat accumulation	Fall et al. [35]
Water-Based Mitigation Strategies					
Urban Water Bodies	Lakes, rivers, ponds, and other natural or artificial water bodies	Areas around water bodies are significantly cooler than other urban areas; forms noticeable cool island effects	Requires adequate water sources; requires regular water quality management	Increases landscape value; provides recreational space; increases air humidity; enhances biodiversity	Ambinakudige (2011) [85], Alhawiti and Mitsova [93]
Fountains and Misting Systems	Installing fountains and misting systems in public spaces to increase evaporative cooling	Can cool local areas by 1–3 °C; more pronounced cooling effect in hot weather	Requires water resources and energy supply; implementation is limited in arid regions; suitable for high-density pedestrian areas	Improves microclimate comfort; increases attractiveness of public spaces; increases air humidity	Du et al. [116]

Table 7. Cont.

Mitigation Strategy	Specific Measures	Cooling Effect	Implementation and Limitations	Conditions	Socioeconomic Benefits	References
Comprehensive Mitigation Strategies						
Multi-Objective Optimization Strategies	Integrated application of multiple mitigation measures, achieving optimal configuration through intelligent algorithms	Application of AI optimization framework can reduce air temperature by 0.7–0.9 °C; can reduce implementation costs by 22.2–42.2%	Requires precise urban thermal environment models; requires multi-department collaboration		Reduces implementation costs; improves resource utilization efficiency; maximizes cooling effects	Qi et al. [6]
Urban Morphology Optimization	Optimizing urban thermal environment through planning building density, height, street width, and other morphological elements	Through combined application of urban forestry, green roofs, and high-albedo surfaces, citywide air temperature can be reduced by 0.4 °C	Applicable to new urban area planning or old city renovation; requires interdisciplinary knowledge		Improves urban ventilation environment; increases urban livability; reduces energy consumption	Rosenzweig et al. [36]

6.3. Remote Sensing-Based Heat Island Mitigation Decision Support

Based on the various heat island mitigation strategies summarized in Table 7, we can construct a comprehensive remote sensing data-driven heat island mitigation decision framework. This framework revolves around three core components: thermal environment assessment and diagnosis, mitigation scenario simulation and evaluation, and implementation monitoring and effect evaluation. In practical applications, Qi et al. [6] developed an artificial intelligence decision framework for Sydney, Australia, demonstrating significant effectiveness by reducing urban air temperature by 0.7–0.9 °C through multi-objective optimization while substantially reducing implementation costs by 22.2–42.2%, achieving a win–win situation of environmental and economic benefits.

Such decision frameworks have been successfully applied and validated in multiple cities globally. Rosenzweig et al. [36] innovatively integrated stakeholder opinions with scientific assessment in New York, achieving a significant city-wide air temperature reduction of 0.4 °C through the synergistic application of urban forestry, green roofs, and high-albedo surfaces. Seeberg et al. [62] utilized long-term Landsat data to evaluate Stuttgart’s heat island mitigation policies, validating the actual effect of reducing overall surface heat island intensity by 1.4 °C and providing scientific support for urban climate policies. Asadi et al. [33] employed artificial neural networks in Austin to precisely simulate green roofs’ heat island mitigation potential, confirming their ability to reduce surface temperature by an average of 1.96 °C. Laurenti Errea et al. [117] analyzed 36 years of heat island changes in Vitoria-Gasteiz, Spain, using multi-generational Landsat satellite data; they revealed the long-term stable effect of minimal heat island intensity changes in urban greening action areas. Research by Liu et al. [108] and Yao et al. [105] in Beijing provided systematic solutions for heat island mitigation in Chinese megacities.

To transform remote sensing data into directly implementable planning decisions, researchers have developed various innovative decision support tools. Park et al. [118] developed a comprehensive methodology integrating 3D urban models with remote sensing land surface temperature data to quantify shade effects from buildings and trees; this demonstrated how urban planners can use advanced spatial analysis to design effective heat mitigation solutions. Quattrochi et al. [119] designed an information support system that cleverly integrated thermal infrared remote sensing data, providing a scientific decision-making basis for urban landscape management. Sun et al. [120] developed an enhanced population heat vulnerability index in the Perth metropolitan area, precisely integrating satellite heat island data with community demographic characteristics. Islam et al. [45] developed a heat island mitigation strategy communication platform based on Landsat 8 and 9 data, promoting effective collaboration among stakeholders in Lahore. Ma et al. [121] achieved a breakthrough by integrating Landsat LST, MODIS LST, and land surface model

simulations, generating gapless surface temperature monitoring at 60 m spatial and half-hourly temporal resolutions.

Remote sensing technology exhibits three major trends in heat island mitigation applications: high-resolution and multi-source data fusion, artificial intelligence and deep learning applications, and cloud computing and open platform popularization. New-generation remote sensing platforms like ECOSTRESS provide 70 m fine thermal imaging, while the GOES-R series achieves ultra-high temporal frequency temperature monitoring every 5 min. Pioneering research by Asadi et al. [33] in Austin significantly improved urban thermal environment analysis accuracy and reliability by fusing Landsat 8 TIRS, Sentinel 2A, and LiDAR data.

Remote sensing-supported heat island mitigation applications are expanding to broader domains. Qi et al.'s [6] research precisely predicted heat-related mortality (5.5–6.4%) and heat discomfort risk, providing a scientific basis for public health decision-making. Todeschi et al. [103] developed the Heat-related Elderly Risk Index (HERI), providing precise protection schemes for specific vulnerable populations. Hulley et al. [23] achieved block-level fine thermal vulnerable community identification in Los Angeles, promoting refined and differentiated urban heat management practices.

Despite significant progress in remote sensing technology for heat island research, challenges remain at the technical, application, and policy levels. Ma et al. [121] successfully achieved seamless LST data generation at a 60-m spatial resolution and half-hourly temporal resolution by integrating satellite observations with land surface model simulations, providing a new technical pathway for addressing the spatiotemporal resolution trade-off challenge. Rosenzweig et al. [36] effectively broke through interdisciplinary communication barriers by integrating climate science, urban planning, and multi-stakeholder participation. Looking forward, with continuous technological innovation and expanding application domains, remote sensing technology will play an increasingly crucial role in urban sustainable development and climate adaptation construction, providing strong scientific support for building livable, resilient, and sustainable urban environments.

7. Challenges and Future Directions

As global climate change intensifies and urbanization continues to advance, the importance of urban heat island research is increasingly prominent, simultaneously raising higher demands for remote sensing monitoring technology. Figure 5 systematically presents the main challenges, emerging technological breakthroughs, and future application directions facing urban heat island remote sensing research, demonstrating the technological pathway from current limitations through emerging analytical methods to ultimately achieving broad social value in the context of climate change. The following will discuss the key components of this developmental pathway and their interrelationships in detail.



Figure 5. Technical approaches and application prospects for future remote sensing research on urban heat islands [6,19,21,32,38,103,122–124].

7.1. Limitations of Existing Technologies and Methods

Despite significant progress in urban heat island remote sensing research, it still faces multiple technical limitations and methodological obstacles. These limitations mainly manifest in three aspects: remote sensing platforms and sensor performance, urban environmental complexity, and methodological standardization, collectively affecting the accuracy and comprehensiveness of heat island monitoring.

Remote sensing platforms and sensor performance constitute the primary factors limiting heat island monitoring effectiveness. The intrinsic balance challenge between geographic detail and observation frequency forms the core issue, as current mainstream remote sensing platforms cannot simultaneously meet the requirements for high spatial resolution and high temporal frequency. The Landsat series provides a 30–100 m spatial resolution, but its 16-day revisit cycle cannot capture rapid changes in urban thermal environments; although MODIS offers higher temporal resolution (1–2 daily observations), its 1 km spatial resolution has difficulty capturing fine thermal variations within cities. This trade-off makes it difficult for researchers to obtain comprehensive thermal environment data, limiting a deeper understanding of urban heat island phenomena. View angle effects and thermal anisotropy further increase monitoring difficulty, with Du et al.'s [19] research showing that using LST observations at sensor zenith angles of $\pm 60^\circ$ can lead to the underestimation of urban surface sensible heat flux and heat island intensity by 45.4% and 43.0%, respectively. Additionally, atmospheric correction complexity significantly affects surface temperature extraction accuracy, particularly in urban areas with high aerosol concentrations, where insufficient atmospheric correction may lead to significant temperature retrieval errors. Cloud cover issues further limit data continuity, with Diem et al. [37] noting that cloud coverage not only leads to data gaps but also affects surface temperature retrieval accuracy, particularly in areas with frequent cloud cover.

Urban environmental complexity constitutes the second major challenge for heat island remote sensing monitoring. The high heterogeneity of urban surface materials

and structures creates difficulties in temperature retrieval processes, with mixtures of different building materials, road surfaces, vegetation, and water bodies within single pixels leading to “mixed pixel” problems; this increases temperature extraction uncertainty, which is particularly evident in medium-to-low resolution sensor data. Simultaneously, the influence of three-dimensional urban structures on thermal radiation is difficult to quantify accurately, as building height, density, and street canyon geometry significantly affect satellite-measured thermal radiation, while traditional two-dimensional remote sensing cannot fully capture these three-dimensional effects. Especially in high-density building areas, urban canyon effects prevent direct satellite observation of some surface temperatures, potentially leading to systematic underestimation of heat island intensity. Furthermore, shadow effects dynamically change over time, and this is particularly evident in high-resolution images, while anthropogenic heat release (such as traffic, industrial activities, and air conditioning system emissions) is difficult to measure directly through traditional remote sensing techniques, with these factors collectively affecting heat island monitoring accuracy.

Difficulties in methodological standardization and data integration constitute the third category of challenges facing heat island research. Inconsistent reference area definitions lead to significant differences in heat island intensity estimation results, with Li et al.’s [125] research indicating that urban-based heat island intensity estimates have obvious uncertainty, primarily stemming from differences in reference area definitions. Difficulties in temporal and spatial scale coordination also affect data integration effectiveness, as scale conversion processes for data from different sensors may lead to information loss or the misinterpretation of heat island patterns. Additionally, data validation challenges significantly limit the assessment of remote sensing result reliability, as ground measurement networks often cannot spatially match satellite observation coverage, with ground truth data scarcity affecting validation result representativeness and reliability. These methodological issues collectively constrain the accuracy, comparability, and application value of heat island research results.

7.2. Application Prospects of Artificial Intelligence and Emerging Analytical Methods

Artificial intelligence and emerging analytical methods provide new pathways for overcoming traditional limitations in remote sensing heat island research, demonstrating broad application prospects. These technological innovations have achieved breakthrough progress, mainly in three aspects: data processing and fusion, thermal environment prediction and simulation, and intelligent decision support; this provides new approaches for high-precision monitoring and the analysis of urban heat islands.

In remote sensing data processing and fusion, artificial intelligence technology has significantly improved heat island monitoring accuracy and continuity. High-resolution thermal data downscaling technology has achieved significant thermal resolution improvement through machine learning methods, with Yao et al. [126] using random forest algorithms to downscale Landsat surface temperature data from 100–120 m to 30 m resolution, greatly enhancing spatial detail in thermal environment analysis. Multi-sensor data fusion has achieved qualitative leaps through artificial intelligence methods, with Adeniran et al. [122] combining multiple satellite data from Landsat, Sentinel, and Himawari, overcoming single-platform spatial-temporal resolution trade-off limitations. Data discontinuity problems caused by cloud coverage have been effectively resolved through machine learning model applications, as these models predict surface temperatures during missing periods by learning temperature spatial-temporal patterns, significantly improving monitoring temporal continuity. Meanwhile, intelligent feature extraction technology has greatly improved surface type recognition accuracy, with Francini et al. [127] using U-Net deep

learning models for building and vegetation segmentation; this provides a solid foundation for studying relationships between urban morphology and thermal environment.

Thermal environment prediction and simulation capabilities have been significantly enhanced through artificial intelligence technology applications. Precise temperature prediction is a core requirement for heat island research, with machine learning models combining remote sensing data and auxiliary variables. Adeniran et al.'s [122] federated learning artificial neural network model achieved high-precision air temperature prediction, with $r = 0.98$ and $RMSE = 0.97$ K, providing important support for heat island risk assessment and warning systems.

Intelligent decision support systems for heat island management represent another important application direction for artificial intelligence technology. The precise identification of thermal vulnerability zones is achieved through integrating remote sensing thermal data with socioeconomic indicators, with Sun et al.'s [120] enhanced population heat vulnerability index providing a scientific basis for targeted heat island mitigation strategy development. Green space planning optimization has achieved unprecedented precision through artificial intelligence methods, with Qi et al.'s [6] multi-objective optimization method capable of predicting the cooling effects of different green space configurations, supporting optimal green space planning design. The same research team demonstrated that urban air temperatures can be reduced by $0.7\text{--}0.9$ °C while reducing implementation costs by 22.2–42.2% through AI decision framework optimization of heat mitigation strategies. The development of these intelligent decision support systems makes urban heat island mitigation planning and implementation more precise and efficient, providing powerful tools for urban climate adaptability.

7.3. Prospects for Urban Heat Island Monitoring in the Context of Climate Change

Against the backdrop of intensifying global climate change, urban heat island monitoring faces new challenges and opportunities. Future research will focus more on the interaction between urban heat islands and climate change, the construction of climate adaptation-oriented monitoring systems, and the integration of emerging observation technologies, collectively promoting urban heat island monitoring toward more refined and comprehensive development.

The interaction between urban heat islands and climate change is an important focus for future research. Rosenzweig et al. [123] noted that under different climate change scenarios, urban heat island effects may exhibit different change trends, with global warming potentially amplifying urban heat island effects and altering thermal difference patterns between cities and surrounding environments. The impact of extreme heat events on urban thermal environments has become a research hotspot, with García [26] using Sentinel-3 data to find that heat island effects may change significantly during heat waves; Zoran et al. [91] also observed that Bucharest's heat island intensity is strongest during summer heat waves. Additionally, changes in precipitation patterns caused by climate change significantly affect urban surface moisture conditions, thereby influencing evaporative cooling and heat island intensity. These complex relationships require combining thermal infrared remote sensing with microwave and optical remote sensing to develop multi-source data fusion methods to achieve comprehensive monitoring of temperature-humidity coupling processes; this would provide a scientific basis for understanding urban heat island evolution patterns against the backdrop of climate change.

Climate adaptation-oriented urban heat island monitoring systems are key to addressing future challenges. Long-term monitoring network construction forms the foundation for understanding climate change impacts, with Ravanelli et al. [128] demonstrating methods for long-term heat island analysis using Google Earth Engine to process historical

Landsat data. Constructing multi-level monitoring systems, including satellite remote sensing, ground meteorological station networks, and citizen science observations, is crucial for discovering long-term interactions between heat island effects and climate change. Multi-scale monitoring strategies support climate adaptation decision-making at different levels, with Kousis et al.'s [38] mobile monitoring system combined with satellite observations constructing cross-scale heat island monitoring networks. Early warning systems are significantly important for heat island and climate change synergistic risk prevention and control. Combining remote sensing real-time monitoring, numerical weather prediction, and artificial intelligence prediction models can help predict and warn of urban heat risks during extreme weather events such as heat waves, providing scientific support for public health emergency response and vulnerable population protection.

The integration of emerging technologies and observation strategies will comprehensively enhance urban heat island monitoring capabilities. New-generation Earth observation satellites, such as Landsat 8/9 and the Sentinel series, have multiple advantages over traditional systems, with Zhou et al. [15] noting exponential growth in heat island research using satellite remote sensing since 2005. UAVs and IoT technologies provide new approaches for urban microclimate monitoring, with Dimitrov et al.'s [129] research demonstrating the value of UAV systems in filling in the scale gaps between satellite remote sensing and ground observations. Citizen science and crowdsourced data collection provide new perspectives for urban heat island research, with temperature data collected by citizen scientists becoming important supplementary data sources as smartphones and portable sensors become more widespread. Digital twin technology provides new platforms for comprehensive urban thermal environment analysis, with Liu and Fan [130] proposing the integration of remote sensing, GIS, and machine learning algorithms to construct urban thermal environment digital twin systems, enabling real-time monitoring, simulation prediction, and scenario analysis. The integrated application of these emerging technologies will greatly enhance the accuracy, comprehensiveness, and real-time nature of urban heat island monitoring, providing stronger scientific support for urban climate adaptation planning.

In summary, future urban heat island remote sensing research will develop along the path of overcoming traditional limitations, applying emerging technologies, and addressing climate change challenges. Through technological innovations such as multi-source data fusion, high-precision thermal environment prediction, and three-dimensional simulation, combined with new-generation satellites, UAVs, digital twins, and other observation methods, urban heat island monitoring will play an increasingly important role in precise heat island risk management and urban climate adaptation planning, providing a scientific basis for urban sustainable development and climate change adaptation.

8. Conclusions

This study systematically reviewed and evaluated the application and development of remote sensing technology in urban heat island effect monitoring and assessment, revealing the technological evolution trajectory from traditional remote sensing platforms to the latest satellite systems, along with accompanying methodological innovations and practical application progress. Through comprehensive multi-dimensional analysis, this study draws the following main conclusions:

First, regarding technological evolution, remote sensing platforms have progressed from early Landsat and NOAA-AVHRR to new-generation systems with higher spatiotemporal resolutions, such as those of the Sentinel series and ECOSTRESS. This technological advancement has greatly expanded the breadth and precision of heat island observation, transforming research from simple static observation to a multi-dimensional, high-precision

comprehensive analysis of urban thermal environments. Particularly, emerging data fusion technologies have successfully overcome the inherent limitations of single sensors in spatiotemporal resolutions, providing more comprehensive and continuous observational data support for heat island research.

Second, at the methodological level, surface temperature extraction algorithms have evolved from Mono-Window Algorithm (MWA) and Split-Window Algorithm (SWA) to specialized algorithms developed for urban environmental complexity, gradually improving temperature retrieval accuracy and adaptability. Multi-source remote sensing data integration and fusion technologies, especially applications of spatiotemporal fusion, down-scaling, and machine learning methods, have effectively overcome the limitations of single data sources, significantly enhancing the comprehensiveness and accuracy of heat island monitoring. A systematic accuracy assessment and validation strategy framework, including ground truth validation, cross-platform validation, and temporal validation methods, provides important guarantees for ensuring the scientific reliability of research results.

Third, regarding heat island characteristics research, this study reveals the nonlinear relationship between city size and heat island intensity by comparing heat island differences between metropolitan areas and small-to-medium cities. Heat island intensity increases by approximately 0.7 °C for each doubling of urban area [30]. Research on the relationships between urban morphology and thermal environment identifies impervious surfaces as the main driver of heat island intensity, explaining over 60% of surface temperature variance in multiple studies, while vegetation shows negative correlations of -0.14 to -0.66 with surface temperature [5], exhibiting significant cooling effects. The analysis of regional differences and universal patterns reveals that, although urban heat island manifestations vary across different climate regions, the proportion of impervious surfaces, vegetation cooling effects, and the positive correlation between urban scale and heat island intensity have demonstrated strong consistency in existing studies.

Fourth, in terms of application value, remote sensing technology provides a scientific basis for precise thermal vulnerability zone identification and green space planning, supports urban material and surface property optimization, and promotes practical effectiveness through heat island mitigation decision support systems. Research shows that urban green spaces can achieve a general cooling of 1–3 °C, locally reaching 5–7 °C [124]; high-albedo materials can reduce surface temperatures by up to 4.5 °C [34]; and AI-optimized heat island mitigation strategies can achieve simultaneous environmental and economic benefits, such as in Sydney, where the urban air temperature was reduced by 0.7–0.9 °C while implementation costs decreased by 22.2–42.2 % [6].

Finally, in facing the limitations of existing technologies and methods—such as inherent trade-offs between spatial and temporal resolution, urban environmental complexity, and methodological standardization difficulties—this study prospects the application potential of artificial intelligence and emerging analytical methods, as well as urban heat island monitoring trends in the context of climate change. Future research should focus on multi-source data fusion, high-precision thermal environment prediction, multi-scale monitoring network construction, and heat island-extreme weather warning system development to support urban climate adaptation planning and precise heat island risk management.

In conclusion, this study constructs a complete knowledge system of “Technology-Method-Application” through a systematic review of the technical methods, research findings, and application practices in urban heat island remote sensing monitoring and assessment. This not only enriches the theoretical foundation of urban climatology but also provides scientific evidence for urban planners, helping to formulate more effective heat island mitigation strategies, enhance urban climate resilience, and provide technical support for achieving the United Nations Sustainable Development Goals related to sustainable

cities and climate action. Future research should focus on multi-source data fusion, high-precision thermal environment prediction, multi-scale monitoring network construction, and heat island–extreme weather early warning system development to support urban climate adaptation planning and precise heat island risk management.

8.1. Critical Knowledge Gaps and Research Challenges

Although urban heat island remote sensing research has made significant progress, several critical knowledge gaps constraining the scientific rigor and effectiveness of heat island monitoring and mitigation still need to be addressed.

8.1.1. Most Urgent Research Challenges

The spatiotemporal resolution trade-off remains fundamentally unresolved. Despite advances in multi-source data fusion technologies, current mainstream remote sensing platforms still cannot simultaneously meet the demands of high spatial resolution and high temporal frequency. The Landsat series provides a 30–100 m spatial resolution, but its 16-day revisit cycle cannot capture rapid changes in urban thermal environments; MODIS has a high temporal resolution (1–2 observations daily), yet its 1-km spatial resolution is inadequate for capturing subtle thermal variations within cities. This trade-off relationship makes it difficult for researchers to obtain comprehensive thermal environment data, limiting an in-depth understanding of urban heat island phenomena.

Three-dimensional urban thermal environment modeling faces major technical bottlenecks. The high heterogeneity of urban surface materials and structures, as well as the influence of three-dimensional urban structures on thermal radiation, are difficult to accurately quantify. Building height, density, and street canyon geometric configurations significantly affect satellite-measured thermal radiation, while traditional two-dimensional remote sensing cannot adequately capture these three-dimensional effects. Particularly in high-density building areas, urban canyon effects prevent the direct satellite observation of some surface temperatures, potentially leading to the systematic underestimation of heat island intensity.

Methodological standardization difficulties constrain research comparability. Inconsistent reference area definitions lead to significant differences in heat island intensity estimation results, and the diverse UHI estimation methods used by different studies make cross-city or cross-study comparisons extremely challenging. Degefu et al. [18] noted that among 263 European cities studied, 11 UHI assessment techniques yielded results with weak negative correlations, further highlighting the urgent need for standardization.

The understanding of heat island–health risk mechanisms remains insufficient. Although research has confirmed that the heat island effect can significantly increase mortality risk during heat waves, e.g., Ho et al. [131] found that high heat island intensity areas showed almost double the mortality risk compared to moderate intensity areas in Hong Kong, the quantitative relationship models between land surface temperature and actual human thermal sensation remain imperfect. In particular, the heat island health risk thresholds across different socioeconomic backgrounds and age groups require in-depth investigation.

8.1.2. Unresolved Technical Limitations

Sensor performance and accuracy limitations. Viewing angle effects and thermal anisotropy are important technical challenges, with Du et al. [19] showing that satellite viewing angle effects can lead to heat island intensity underestimation by up to 45.4%. Additionally, the complexity of atmospheric correction significantly affects land surface temperature extraction accuracy, particularly in urban areas with high aerosol concentrations, where inadequate atmospheric correction may cause significant temperature retrieval errors.

Data acquisition and environmental factor limitations. Cloud cover severely affects the availability and quality of remote sensing data, with Degefu et al. [18] showing that among 21 images browsed in the United States, only one image was cloud-free. Urban areas, due to strong convective activities, have higher probabilities of afternoon cloud formation, further limiting data acquisition frequency.

Incomplete validation and accuracy assessment systems. Ground measurement networks often cannot match satellite observation coverage in spatial distribution, and the scarcity of ground truth data affects the representativeness and reliability of validation results. Degefu et al. [18] emphasized that many studies using Landsat and MODIS did not report the accuracy of retrieved LST, limiting the accurate assessment of remote sensing UHI research reliability.

Mixed pixel and urban complexity challenges. The mixing of different building materials, road surfaces, vegetation, and water bodies within single pixels creates “mixed pixel” problems, increasing uncertainty in temperature extraction; this is particularly evident in medium-to-low-resolution sensor data. Meanwhile, shadow effects change dynamically over time, and this is especially noticeable in high-resolution images, and anthropogenic heat releases (such as traffic, industrial activities, and air conditioning system emissions) are difficult to measure directly through traditional remote sensing techniques.

8.2. Promising Directions for Future Investigation and Policy Development

8.2.1. Key Directions for Technical Development

The integration and application of new-generation remote sensing technologies. New-generation Earth observation satellites such as Landsat 8/9 and the Sentinel series have significant advantages, with Zhou et al. [15] noting that heat island research using satellite remote sensing has grown exponentially since 2005. ECOSTRESS represents an important breakthrough in spatial thermal imaging, with its 70 m resolution achieving an order-of-magnitude improvement compared to the 1 km resolution of MODIS; this enables the clear distinction of the thermal characteristics of individual urban elements such as buildings, parking lots, and small green spaces.

Artificial intelligence-driven intelligent heat island analysis. Artificial intelligence technology shows broad prospects in data processing and fusion, thermal environment prediction and simulation, and intelligent decision support. Yao et al. [126] used random forest algorithms to downsample Landsat land surface temperature data from 100–120 m to 30 m resolution, significantly improving spatial details in thermal environment analysis. Machine learning methods, with their powerful nonlinear modeling capabilities, demonstrate significant advantages in multi-source data fusion.

Deepening multi-source data fusion technologies. Multi-source data integration technologies such as spatiotemporal fusion, downscaling techniques, and machine learning methods effectively overcome the limitations of single data sources. Bird et al. [24] combined Landsat and MODIS data to achieve temperature monitoring at a 100-m spatial resolution and a daily temporal resolution, demonstrating the application value of data fusion technologies.

8.2.2. Key Directions for Future Research and Policy Development

The related scientific research frontiers show three main development trends. The interaction between urban heat islands and climate change has become an important focus for future research. García [26] used Sentinel-3 data to discover significant changes in heat island effects during heat waves, and Zoran et al. [91] observed enhanced heat island intensity in Bucharest during summer heat waves, indicating the important impact of extreme climate events on urban thermal environments. This coupled-mechanism research

will provide scientific evidence for understanding urban thermal environment evolution under climate change. Multi-scale heat island monitoring system construction is equally crucial, as the multi-scale characteristics of urban heat islands require corresponding spatial resolutions and technical methods at different planning levels. Cross-scale data fusion and scale conversion from the urban regional scale to the microclimate scale represent important approaches to address single-scale limitations. Additionally, social–environmental coupled heat island research is becoming an emerging research direction, with heat vulnerability identification requiring the integration of remote sensing thermal data with socioeconomic indicators. Chakraborty et al.'s [7] study of 497 US urban areas quantified environmental inequality phenomena, providing important directions for environmental justice-oriented heat island research.

At the policy formulation and implementation level, future development needs to advance under four key directions. The construction of a standardized system is fundamental work, requiring the establishment of unified heat island intensity calculation standards, reference area definition specifications, and data quality control systems to improve comparability between different studies and the scientific value of heat island data. Improving multi-level governance systems requires fully leveraging the important role of remote sensing technology in heat vulnerability identification, green space planning, and urban material optimization, comprehensively integrating remote sensing heat island monitoring results in urban planning, and establishing evidence-based heat island mitigation decision support systems. Considering that existing research is mainly concentrated in developed countries and regions with uneven geographical coverage, international cooperation and capacity building are particularly important. International cooperation should be strengthened to promote heat island monitoring technology transfer to developing countries and narrow global heat island research technology gaps. Finally, public health-oriented heat island management should become a policy priority, with recommendations to establish urban thermal health risk assessment and early warning systems based on remote sensing data, prioritize the protection of vulnerable groups, such as the elderly and children, and achieve precise heat island health protection.

8.3. Review Value and Prospects

This review provides comprehensive theoretical foundations and practical guidance for urban heat island remote sensing research through the systematic construction of a “Technology-Method-Application” knowledge framework based on an in-depth analysis of the high-quality literature. In terms of theoretical contributions, it clarifies the technological evolution patterns of heat island remote sensing monitoring from early traditional platforms to new-generation systems, systematically organizes the methodological development trajectory from mono-window algorithms to urban-specific algorithms, and identifies application expansion pathways from heat vulnerability identification to intelligent decision support.

In terms of practical value, it provides scientific evidence for urban planners regarding remote sensing-based heat island mitigation, offers decision-support frameworks for policymakers in climate adaptation planning, and identifies key directions for researchers in technological innovations such as multi-source data fusion and artificial intelligence applications. The universal cooling effects of urban green spaces (1–3 °C), the cooling potential of high-albedo materials (up to 4.5 °C), and the cost–benefit optimization achieved through AI-optimized strategies provide quantified scientific support for heat island mitigation practices.

Looking forward, urban heat island remote sensing research will achieve important developments at three levels: technological innovation, methodological improvement,

and application expansion. At the technical level, new-generation sensors and multi-source data fusion will significantly enhance monitoring accuracy; at the methodological level, artificial intelligence technology will enable intelligent analysis and the prediction of thermal environments; at the application level, precise health protection and climate adaptation planning will become important directions for heat island governance.

Under the dual challenges of global climate change and rapid urbanization, the social value of urban heat island remote sensing research will become more prominent. Through continuous technological innovation, methodological improvement, and application expansion, remote sensing technology will provide stronger technological support for building climate-resilient cities, ensuring residents' health and welfare, and achieving sustainable development goals. It is hoped that this review will promote in-depth research and widespread application in related fields, contributing scientific wisdom to addressing global urban thermal environment challenges.

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